

07_final_figures

June 24, 2026

1 Step 7: Final Results Synthesis and Paper Figure Selection

This notebook assembles the manuscript-facing figure and table set from the authoritative v4 outputs. It focuses on empirical evidence and final reporting choices: which existing step-specific figures are strongest for the main paper, which belong in the supplement, and which tables should carry exact reported values.

The study-design schematic will be handled directly in the manuscript source, so this notebook concentrates on the finalized analytical figures and tables.

```
[24]: import sys
from pathlib import Path

import pandas as pd
from IPython.display import Image, display

def find_repo_root() -> Path:
    candidates = [Path.cwd().resolve(), *Path.cwd().resolve().parents]
    for base in candidates:
        if (base / "data" / "processed" / "v4" / "v4_protocol_manifest.json").
        exists():
            return base
    raise RuntimeError("Could not locate repository root from the current_
    working directory.")

REPO_ROOT = find_repo_root()
if str(REPO_ROOT) not in sys.path:
    sys.path.insert(0, str(REPO_ROOT))
if str(REPO_ROOT / "src") not in sys.path:
    sys.path.insert(0, str(REPO_ROOT / "src"))

FIG_PDF_ROOT = REPO_ROOT / "report" / "figures" / "v4" / "pdf"
FIG_PNG_ROOT = REPO_ROOT / "report" / "figures" / "v4" / "png"
TABLE_DIR = REPO_ROOT / "report" / "tables" / "v4"
TABLE_DIR.mkdir(parents=True, exist_ok=True)
```

```

DIRECTIONS = ["pd_to_hd", "hd_to_pd", "pd_to_als", "als_to_pd", "hd_to_als",
↪ "als_to_hd"]
DIR_LABEL = {d: d.replace("_to_", "→").upper() for d in DIRECTIONS}
CLF_LABEL = {
    "rf": "RF",
    "knn": "KNN",
    "svm": "SVM",
    "dt": "DT",
    "qda": "QDA",
    "xgb": "XGBoost",
    "lgbm": "LightGBM",
}

def figure_paths(step: str, stem: str) -> tuple[Path, Path]:
    return FIG_PDF_ROOT / step / f"{stem}.pdf", FIG_PNG_ROOT / step / f"{stem}.
↪ png"

def display_existing_figure(step: str, stem: str, width: int = 780) -> None:
    pdf_path, png_path = figure_paths(step, stem)
    assert png_path.exists(), f"Missing PNG figure: {png_path}"
    assert pdf_path.exists(), f"Missing PDF figure: {pdf_path}"
    print(f"Displayed figure: {png_path.relative_to(REPO_ROOT)}")
    print(f"Vector source: {pdf_path.relative_to(REPO_ROOT)}")
    display(Image(filename=png_path, width=width))

def save_table_csv(df: pd.DataFrame, name: str) -> Path:
    path = TABLE_DIR / name
    df.to_csv(path, index=False)
    print(f"Saved table: {path.relative_to(REPO_ROOT)}")
    return path

def save_table_latex(
    df: pd.DataFrame,
    name: str,
    caption: str,
    label: str,
    *,
    longtable: bool = False,
) -> Path:
    path = TABLE_DIR / name
    latex = df.to_latex(
        index=False,
        escape=False,

```

```

        caption=caption,
        label=label,
        longtable=longtable,
    )
    path.write_text(latex)
    print(f"Saved table: {path.relative_to(REPO_ROOT)}")
    return path

```

```

FIGURE_SPECS = [
    {
        "tier": "main",
        "step": "step0",
        "figure_stem": "step0_authoritative_filtering_cascade",
        "recommended_caption": "Authoritative v4 filtering cascade from raw_
↪stride rows to the retained cohort.",
        "claim_supported": "The retained dataset is auditable from raw rows to_
↪the frozen analytical matrix.",
        "why_included": "Cohort accounting is essential for a publication-track_
↪biomedical ML study.",
        "caveat": "Descriptive only; not a performance result.",
    },
    {
        "tier": "main",
        "step": "step1",
        "figure_stem": "step1_control_split_summary",
        "recommended_caption": "Frozen Control A / Control B split used for_
↪transfer and control-partition sensitivity analyses.",
        "claim_supported": "Downstream transfer and Step 8 sensitivity rely on_
↪an explicit disjoint-control design.",
        "why_included": "This is the clearest visual anchor for the_
↪control-partition methodology.",
        "caveat": "Small cohorts still permit sensitivity to alternate_
↪near-optimal control splits.",
    },
    {
        "tier": "main",
        "step": "step2",
        "figure_stem": "f1_within_condition_heatmap",
        "recommended_caption": "Subject-level within-condition macro-F1 across_
↪classifiers and source conditions.",
        "claim_supported": "The source-condition benchmark is strong enough to_
↪make downstream transfer interpretable.",
        "why_included": "This is the cleanest subject-primary overview of Step_
↪2.",
    },

```

```

    "caveat": "ALS still requires small-sample caution despite its strong
↪scores.",
  },
  {
    "tier": "main",
    "step": "step2",
    "figure_stem": "step2_subject_stride_leaders",
    "recommended_caption": "Condition-level leaders with subject-primary
↪and stride-level companion metrics.",
    "claim_supported": "The paper can report exact Step 2 leaders without
↪hard-coded manuscript values.",
    "why_included": "The panel complements the heatmap with exact
↪condition-level numbers.",
    "caveat": "Nested winners are protocol-specific, not universal
↪classifier rankings.",
  },
  {
    "tier": "main",
    "step": "step3",
    "figure_stem": "degradation_heatmap",
    "recommended_caption": "Subject-level matched degradation by transfer
↪direction and classifier.",
    "claim_supported": "Transfer is measurable but usually degraded
↪relative to within-source subject baselines.",
    "why_included": "Matched degradation is the central Step 3 comparison
↪lens.",
    "caveat": "Direction-level means are uniformly positive, but individual
↪classifier-direction cells still vary.",
  },
  {
    "tier": "main",
    "step": "step3",
    "figure_stem": "cross_condition_f1_heatmap",
    "recommended_caption": "Absolute subject-level zero-shot transfer
↪performance by direction and classifier.",
    "claim_supported": "Transfer success is directional and
↪classifier-dependent rather than uniform.",
    "why_included": "It complements matched degradation with the absolute
↪transfer-performance surface.",
    "caveat": "Absolute F1 should still be interpreted alongside the
↪within-source baseline.",
  },
  {
    "tier": "main",
    "step": "step3",
    "figure_stem": "step3_subject_confusion_matrices",

```

```

    "recommended_caption": "Subject-level normalized confusion matrices for
↳the best classifier in each transfer direction.",
    "claim_supported": "The strongest transfer directions remain
↳interpretable in terms of control specificity and target-disease recall.",
    "why_included": "This panel adds error-structure information that the
↳heatmaps do not show directly.",
    "caveat": "The matrices summarize only the direction-level leaders, not
↳every classifier.",
  },
  {
    "tier": "extended_main",
    "step": "step3",
    "figure_stem": "step3_direction_intervals",
    "recommended_caption": "Direction-level subject-F1 intervals for the
↳best classifier in each transfer direction.",
    "claim_supported": "The strongest transfer directions still carry
↳non-trivial uncertainty.",
    "why_included": "Useful when the manuscript wants explicit interval
↳context beyond heatmaps.",
    "caveat": "Descriptive intervals only; not a primary direction-level
↳inferential test.",
  },
  {
    "tier": "extended_main",
    "step": "step3",
    "figure_stem": "step3_bidirectional_asymmetry",
    "recommended_caption": "Bidirectional asymmetry of subject-level
↳transfer between disease pairs.",
    "claim_supported": "Source-to-target direction matters and cannot be
↳treated as interchangeable.",
    "why_included": "This is the most direct visualization of the asymmetry
↳claim.",
    "caveat": "Asymmetry should be interpreted within this repaired v4
↳protocol, not as a universal disease ordering.",
  },
  {
    "tier": "main",
    "step": "step4",
    "figure_stem": "step4_delta_j_rf_heatmap",
    "recommended_caption": "Random-forest SHAP reliance shifts under
↳transfer.",
    "claim_supported": "Transfer alters feature reliance in concrete,
↳direction-specific ways for a representative strong model family.",
    "why_included": "It provides a model-specific diagnostic anchor for the
↳broader SHAP story.",
  }

```

```

    "caveat": "Model-specific SHAP movement should not be generalized_
↪without the consensus views.",
  },
  {
    "tier": "main",
    "step": "step4",
    "figure_stem": "step4_delta_j_normalized_consensus_heatmap",
    "recommended_caption": "Normalized consensus SHAP reliance shifts_
↪across classifiers and transfer directions.",
    "claim_supported": "Several timing-structure features recur as_
↪transfer-shift signals across model families.",
    "why_included": "This is the strongest cross-classifier Step 4 panel.",
    "caveat": "Diagnostic only; not evidence of causal gait biomarkers.",
  },
  {
    "tier": "main",
    "step": "step4",
    "figure_stem": "step4_delta_j_spearman",
    "recommended_caption": "Direction-wise cross-classifier rank agreement_
↪of SHAP transfer-shift features.",
    "claim_supported": "Some directions show coherent diagnostic ranking_
↪across classifier families, whereas others are more fragmented.",
    "why_included": "It indicates when the transfer-diagnosis story is_
↪shared across models rather than model-specific.",
    "caveat": "Agreement patterns are descriptive and should not be_
↪overread as proof of a single latent mechanism.",
  },
  {
    "tier": "extended_main",
    "step": "step4",
    "figure_stem": "step4_family_delta_j_consensus",
    "recommended_caption": "Family-level consensus movement in feature_
↪reliance under transfer.",
    "claim_supported": "Variability and raw timing dominate the_
↪family-level transfer-diagnostic story.",
    "why_included": "This is the cleanest abstraction over the_
↪feature-level maps.",
    "caveat": "Family movement summarizes reliance shifts, not biology.",
  },
  {
    "tier": "extended_main",
    "step": "step4",
    "figure_stem": "step4_top_feature_recurrence",
    "recommended_caption": "Recurring top-ranked transfer-shift features_
↪across directions and classifiers.",
  }

```

```

    "claim_supported": "A small set of timing-variability descriptors
↳repeatedly reappears in the transfer-diagnostic layer.",
    "why_included": "Useful as a feature-level complement to the family
↳summary.",
    "caveat": "Recurrence is descriptive rather than causal.",
  },
  {
    "tier": "main",
    "step": "step5",
    "figure_stem": "noise_curves_cross",
    "recommended_caption": "Cross-condition Gaussian feature-space stress
↳curves across transfer directions.",
    "claim_supported": "Cross-condition performance deteriorates under
↳increasing feature-space stress in a direction-dependent way.",
    "why_included": "This is the most compact robustness summary for
↳transferred models.",
    "caveat": "Gaussian feature-space stress is not raw sensor simulation.",
  },
  {
    "tier": "main",
    "step": "step5",
    "figure_stem": "step5_feature_sensitivity_within",
    "recommended_caption": "Within-condition single-feature permutation
↳sensitivity heatmap.",
    "claim_supported": "Within-condition models show selective reliance
↳patterns that provide context for transfer sensitivity.",
    "why_included": "The within-condition perturbation surface is
↳scientifically distinct from the cross-condition one.",
    "caveat": "Permutation sensitivity is a perturbation diagnostic rather
↳than a causal attribution.",
  },
  {
    "tier": "main",
    "step": "step5",
    "figure_stem": "step5_feature_sensitivity_cross",
    "recommended_caption": "Cross-condition single-feature permutation
↳sensitivity heatmap.",
    "claim_supported": "Transfer performance is especially sensitive to a
↳small set of timing-variability features.",
    "why_included": "This figure links the Step 4 reliance-shift layer to a
↳direct perturbation response surface.",
    "caveat": "Cross-condition sensitivity under permutation does not
↳establish deployment fragility in the real world.",
  },
  {
    "tier": "main",

```

```

    "step": "step5",
    "figure_stem": "conformal_within",
    "recommended_caption": "Within-condition exploratory conformal coverage
↪and set-size behavior.",
    "claim_supported": "Source-calibrated set behavior differs materially
↪by method and alpha even before distribution shift.",
    "why_included": "It gives the matched-domain reference for the
↪conformal diagnostics.",
    "caveat": "Descriptive only; not the primary performance endpoint.",
  },
  {
    "tier": "main",
    "step": "step5",
    "figure_stem": "conformal_cross",
    "recommended_caption": "Cross-condition exploratory conformal coverage
↪and set-size behavior under shift.",
    "claim_supported": "Prediction-set behavior changes under transfer and
↪should be framed as exploratory under domain shift.",
    "why_included": "It directly supports the caution that target-domain
↪coverage is not guaranteed.",
    "caveat": "No formal target-domain conformal guarantee should be
↪claimed here.",
  },
  {
    "tier": "extended_main",
    "step": "step5",
    "figure_stem": "step5_feature_family_sensitivity",
    "recommended_caption": "Feature-family permutation sensitivity across
↪within and transfer settings.",
    "claim_supported": "Variability and raw timing families dominate the
↪average transfer sensitivity story.",
    "why_included": "This is a strong higher-level complement to the
↪feature heatmaps.",
    "caveat": "Family-level averages can hide direction-specific exceptions.
↪",
  },
  {
    "tier": "extended_main",
    "step": "step5",
    "figure_stem": "step5_corruption_cross",
    "recommended_caption": "Structured cross-condition corruption response
↪across severity levels.",
    "claim_supported": "Heavy benchmark label corruption and heavy
↪feature-space jitter are among the most damaging average cross-condition
↪stressors.",

```

```

    "why_included": "Adds a perturbation family that is distinct from
↳ Gaussian stress and feature permutation.",
    "caveat": "These are benchmark perturbations, not deployment-validated
↳ sensor-fault models.",
  },
  {
    "tier": "main",
    "step": "step6",
    "figure_stem": "kmeans_scatter_k3",
    "recommended_caption": "Exploratory K-means overlay in subject-level
↳ PCA space.",
    "claim_supported": "The subject-level feature space has visible
↳ structure, but that structure should remain exploratory rather than
↳ confirmatory.",
    "why_included": "This is the most visually direct Step 6 panel for
↳ qualitative geometry context.",
    "caveat": "Exploratory only; clustering does not prove disease
↳ separability or transfer validity.",
  },
  {
    "tier": "extended_main",
    "step": "step6",
    "figure_stem": "pca_scatter_by_condition",
    "recommended_caption": "Subject-level PCA projection colored by
↳ condition.",
    "claim_supported": "The condition geometry is visually structured but
↳ remains only qualitative context for the supervised results.",
    "why_included": "Useful if the supplement wants the raw PCA view
↳ alongside the K-means overlay.",
    "caveat": "Visual separation alone is not supervised evidence.",
  },
  {
    "tier": "main",
    "step": "step8",
    "figure_stem": "step8_sign_stability_matrix",
    "recommended_caption": "Sign stability of direction-level matched
↳ degradation across near-optimal control partitions.",
    "claim_supported": "Step 8 qualifies stability claims rather than
↳ proving invariance.",
    "why_included": "This is the clearest figure for the key Step 8 caveat.
↳",
    "caveat": "Partition 3 reverses four of six direction-level signs.",
  },
  {
    "tier": "main",
    "step": "step8",

```

```

        "figure_stem": "step8_direction_degradation_ranges",
        "recommended_caption": "Direction-wise matched-degradation ranges
↪across the main and alternate near-optimal control partitions.",
        "claim_supported": "Control-partition sensitivity affects both sign and
↪effect-size scale.",
        "why_included": "It complements the sign matrix with direction-level
↪magnitude movement.",
        "caveat": "Sensitivity includes movement in both within-source
↪baselines and transfer outcomes.",
    },
    {
        "tier": "extended_main",
        "step": "step8",
        "figure_stem": "step8_within_source_baselines",
        "recommended_caption": "Best within-source subject-level baselines
↪across near-optimal control partitions.",
        "claim_supported": "Some Step 8 movement is driven by baseline
↪instability, not transfer behavior alone.",
        "why_included": "Important supporting context for the sensitivity
↪verdict.",
        "caveat": "Baseline drift does not by itself imply transfer failure.",
    },
    {
        "tier": "extended_main",
        "step": "step8",
        "figure_stem": "step8_best_transfer_leaders_heatmap",
        "recommended_caption": "Best transfer leaders by direction and control
↪partition.",
        "claim_supported": "The identity of the best transfer classifier is not
↪completely fixed across alternate partitions.",
        "why_included": "Helpful if the supplement wants the classifier-family
↪stability view explicitly.",
        "caveat": "Best-leader changes should be read alongside effect-size
↪movement, not in isolation.",
    },
]

TABLE_SPECS = [
    {
        "tier": "main",
        "table_file": "step2_subject_stride_leaders_v4.csv",
        "purpose": "Condition-level within-condition leaders with
↪subject-primary and stride-level companions.",
        "main_columns": "Condition, Leader, Subject F1, Stride F1, Subject log
↪loss, Strategy",
        "paper_use": "Exact Step 2 leader reporting.",
    }
]

```

```

    "caveat": "Nested winners are protocol-specific rather than universal_
↪rankings.",
  },
  {
    "tier": "main",
    "table_file": "paper_main_results_summary_v4.csv",
    "purpose": "Full direction-by-classifier cross-condition transfer_
↪summary with subject-primary columns emphasized.",
    "main_columns": "Direction, Classifier, Subject F1, Within-source_
↪subject F1, Matched degradation (subject), Control specificity, Target_
↪recall, Strategy, Permutation p-value",
    "paper_use": "Primary Step 3 numerical reporting table.",
    "caveat": "Direction-level means are the safest headline summaries; not_
↪every classifier-direction row degrades.",
  },
  {
    "tier": "main",
    "table_file": "step5_noise_robustness_summary_v4.csv",
    "purpose": "Sigma-zero replay and key severity checkpoints for the_
↪Gaussian feature-space stress test.",
    "main_columns": "scope, classifier, f1@0, f1@0.10, f1@0.25, f1@0.50,_
↪sigma_at_10pct_drop",
    "paper_use": "Exact Step 5 checkpoint reporting.",
    "caveat": "This is a feature-space perturbation diagnostic, not a raw_
↪sensor benchmark.",
  },
  {
    "tier": "main",
    "table_file": "step8_direction_sensitivity_v4.csv",
    "purpose": "Direction-level matched-degradation sensitivity across_
↪alternate control partitions.",
    "main_columns": "Partition, Direction, Mean matched degradation, Sign,_
↪Best classifier, Best subject F1",
    "paper_use": "Exact Step 8 sensitivity reporting.",
    "caveat": "Sensitivity qualifies the main transfer story rather than_
↪replacing it.",
  },
  {
    "tier": "extended_main",
    "table_file": "step0_filter_flow.csv",
    "purpose": "Cohort filtering stages and retained stride-row counts.",
    "main_columns": "stage_label, rows",
    "paper_use": "Methods transparency for cohort accounting.",
    "caveat": "Descriptive only.",
  },
  {

```

```

        "tier": "extended_main",
        "table_file": "step1_control_partition_summary.csv",
        "purpose": "Frozen Control A / Control B membership.",
        "main_columns": "subject_id, group",
        "paper_use": "Methods or supplement table for control-reuse_
↪transparency.",
        "caveat": "Important for protocol clarity but not a results table.",
    },
    {
        "tier": "extended_main",
        "table_file": "step3_best_direction_recall_summary_v4.csv",
        "purpose": "Best classifier per transfer direction with recall and_
↪specificity context.",
        "main_columns": "Direction, Classifier, Subject F1, Delta subject,
↪Control specificity, Target recall",
        "paper_use": "Compact Step 3 companion table if the manuscript wants_
↪the best-direction subset explicitly.",
        "caveat": "Best-direction rows do not summarize the full classifier_
↪surface.",
    },
    {
        "tier": "extended_main",
        "table_file": "step5_feature_family_sensitivity_v4.csv",
        "purpose": "Family-level permutation sensitivity across within and_
↪transfer settings.",
        "main_columns": "scope, domain, classifier, family, drop",
        "paper_use": "Step 5 companion table for the feature-family story.",
        "caveat": "Family averages compress heterogeneous direction-specific_
↪behavior.",
    },
    {
        "tier": "extended_main",
        "table_file": "step8_partition_overview_v4.csv",
        "purpose": "Partition-level Step 8 overview with overlap and_
↪role-reversal metadata.",
        "main_columns": "partition, role, overlap,
↪independent_sensitivity_candidate, candidate_family",
        "paper_use": "Supplementary context for the control-partition design.",
        "caveat": "Structural metadata rather than a headline results table.",
    },
]

LEGACY_DUPLICATE_TABLES = {
    "step7_within_leaders_v4.csv",
    "step7_within_leaders_v4.tex",
    "step7_cross_transfer_leaders_v4.csv",

```

```

"step7_cross_transfer_leaders_v4.tex",
"step7_master_transfer_table_v4.csv",
"step7_master_transfer_table_v4.tex",
}

```

1.1 Manuscript selection criteria

The manuscript should stay subject-primary, extensive enough to cover every major result layer, and disciplined enough to avoid near-duplicate panels.

- Step-specific figures are preferred over regenerated summary surrogates when the existing figure already states the result clearly.
- Main-paper figures should support distinct claims: cohort accounting, frozen control design, within-condition benchmarking, zero-shot transfer, transfer-diagnostic evidence, stress testing and uncertainty behavior, exploratory geometry context, and control-partition sensitivity.
- Dense diagnostics still matter scientifically, but they move to `extended_main` or `supplement` when they repeat the same claim at finer granularity.

1.2 Methods and cohort figures

The paper's methods/result interface should begin with two concrete anchors: the authoritative filtering cascade and the frozen Control A / Control B partition. Together they make the retained cohort auditable and clarify why the later transfer and sensitivity analyses can be interpreted as leakage-aware.

```

[25]: display_existing_figure("step0", "step0_authoritative_filtering_cascade",
    ↪width=800)
display_existing_figure("step1", "step1_control_split_summary", width=800)

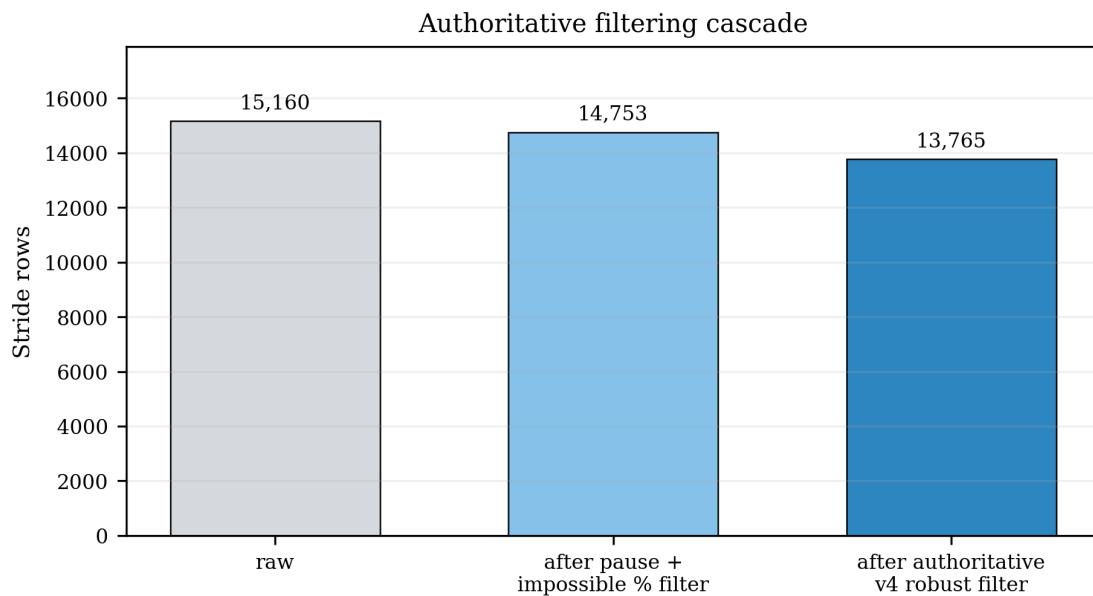
```

Displayed figure:

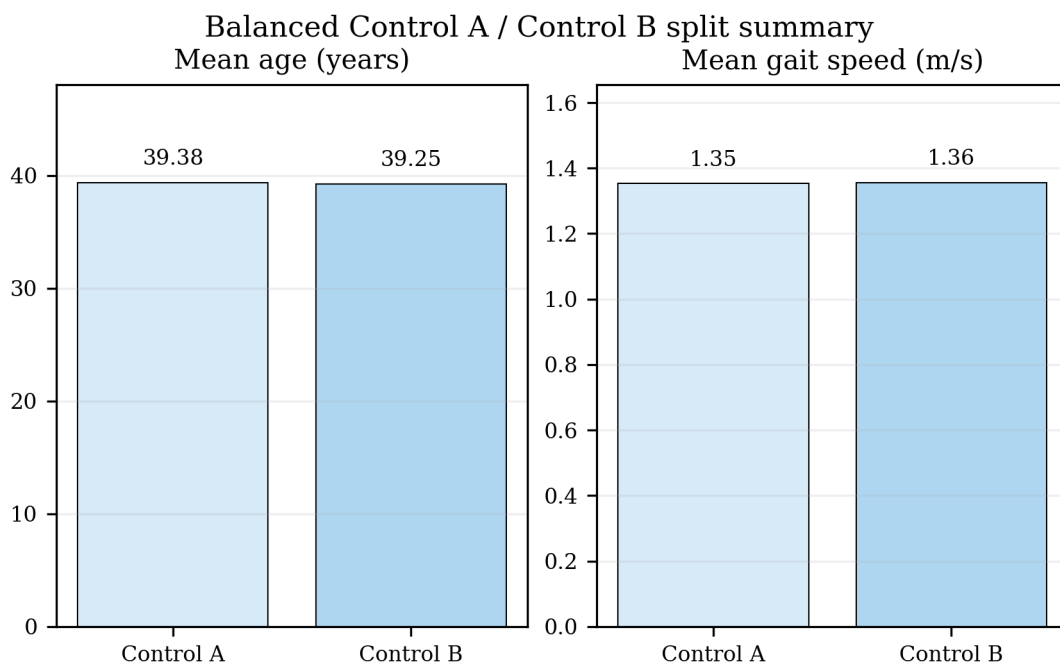
report/figures/v4/png/step0/step0_authoritative_filtering_cascade.png

Vector source:

report/figures/v4/pdf/step0/step0_authoritative_filtering_cascade.pdf



Displayed figure: `report/figures/v4/png/step1/step1_control_split_summary.png`
 Vector source: `report/figures/v4/pdf/step1/step1_control_split_summary.pdf`



- **step0_authoritative_filtering_cascade** is a main paper figure because it shows the retained-row path from **15,160** raw stride rows to **13,765** authoritative v4 rows. It sup-

ports cohort transparency rather than a performance claim.

- **step1_control_split_summary** is also main because the later transfer protocol and Step 8 sensitivity analysis depend on a frozen **8-vs-8** Control A / Control B split. The main caveat is that a balanced split is not the same thing as a partition-invariant result.

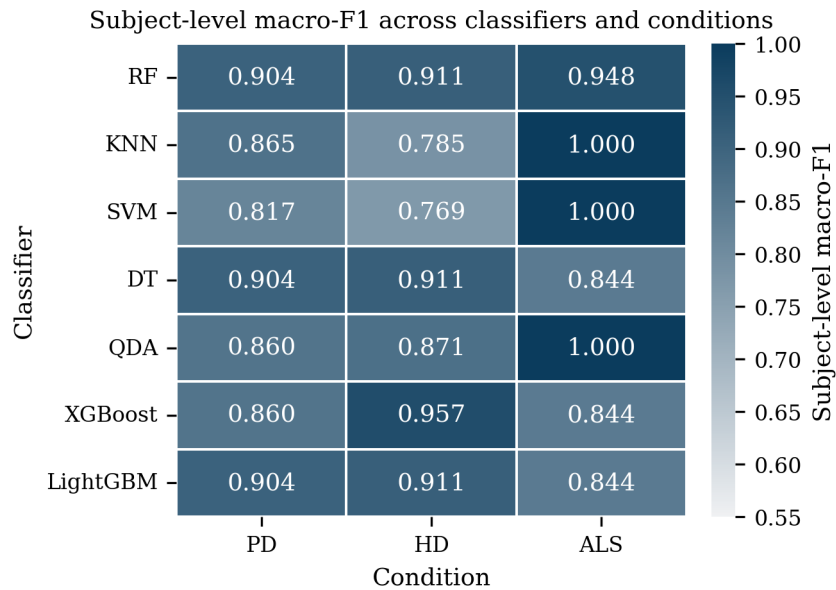
1.3 Step 2: within-condition benchmark

The Step 2 figures establish whether the source-condition models are strong enough to support a downstream transfer analysis. The paper should therefore show both the subject-level leaderboard surface and the compact leader summary.

```
[26]: step2_leaders = pd.read_csv(TABLE_DIR / "step2_subject_stride_leaders_v4.csv")
display_existing_figure("step2", "f1_within_condition_heatmap", width=780)
display_existing_figure("step2", "step2_subject_stride_leaders", width=780)
display(step2_leaders)
```

Displayed figure: report/figures/v4/png/step2/f1_within_condition_heatmap.png

Vector source: report/figures/v4/pdf/step2/f1_within_condition_heatmap.pdf



Displayed figure: report/figures/v4/png/step2/step2_subject_stride_leaders.png

Vector source: report/figures/v4/pdf/step2/step2_subject_stride_leaders.pdf

Subject-primary leaders by condition

Condition	Leader	Subject F1	Stride F1	Subject log loss	Strategy
ALS	QDA	1.0	0.9642	0.0481	synthetic
HD	XGBoost	0.9571	0.9546	0.1524	raw
PD	DT	0.9042	0.9022	1.2065	balanced

	Condition	Leader	Subject F1	Stride F1	Subject log loss	Strategy
0	ALS	QDA	1.0000	0.9642	0.0481	synthetic
1	HD	XGBoost	0.9571	0.9546	0.1524	raw
2	PD	DT	0.9042	0.9022	1.2065	balanced

The subject-primary Step 2 story is crisp. **ALS** is the strongest source condition and reaches a **QDA** leader score of **1.0000** subject-level macro-F1; **HD** is led by **XGBoost** at **0.9571**; and **PD** is flatter, with **DT** selected as the authoritative leader at **0.9042** subject-level macro-F1. That combination matters for Step 3 because it means the transfer analysis starts from strong, but not identical, within-source baselines. The main caveat is that these leaders are optimal within the repaired nested procedure only; they should not be described as universal classifier winners.

1.4 Step 3: zero-shot cross-condition transfer

Step 3 is the core empirical result. The manuscript should keep both the absolute transfer surface and the matched-degradation lens, then pair them with one detailed diagnostic view and one exact-value table so that the main claim does not rest on heatmaps alone.

```
[27]: transfer_df = pd.read_csv(TABLE_DIR / "step3_cross_condition_summary_v4.csv")
transfer_table = transfer_df.copy()
transfer_table["Direction"] = transfer_table["direction"].map(DIR_LABEL)
transfer_table["Classifier"] = transfer_table["classifier"].map(CLF_LABEL)
paper_main_results_summary = transfer_table[
    [
        "Direction",
        "Classifier",
        "subject_f1",
        "within_subject_f1",
        "delta_subject",
        "stride_f1",
        "within_stride_f1",
        "delta_stride",
        "subject_ci_low",
        "subject_ci_high",
    ]
]
```

```

        "control_specificity_subject",
        "target_recall_subject",
        "selected_strategy",
        "permutation_p_value",
    ]
].rename(
    columns={
        "subject_f1": "Subject F1",
        "within_subject_f1": "Within-source subject F1",
        "delta_subject": "Matched degradation (subject)",
        "stride_f1": "Stride F1",
        "within_stride_f1": "Within-source stride F1",
        "delta_stride": "Matched degradation (stride)",
        "subject_ci_low": "Subject CI low",
        "subject_ci_high": "Subject CI high",
        "control_specificity_subject": "Control specificity",
        "target_recall_subject": "Target recall",
        "selected_strategy": "Strategy",
        "permutation_p_value": "Permutation p-value",
    }
)
save_table_csv(paper_main_results_summary, "paper_main_results_summary_v4.csv")
save_table_latex(
    paper_main_results_summary.round(4),
    "paper_main_results_summary_v4.tex",
    "Cross-condition transfer summary with subject-primary columns emphasized.",
    "tab:paper_main_results_summary_v4",
    longtable=True,
)

styled_transfer = (
    paper_main_results_summary.style
    .format({
        "Subject F1": "{:.4f}",
        "Within-source subject F1": "{:.4f}",
        "Matched degradation (subject)": "{:+.4f}",
        "Stride F1": "{:.4f}",
        "Within-source stride F1": "{:.4f}",
        "Matched degradation (stride)": "{:+.4f}",
        "Subject CI low": "{:.4f}",
        "Subject CI high": "{:.4f}",
        "Control specificity": "{:.4f}",
        "Target recall": "{:.4f}",
        "Permutation p-value": "{:.4f}",
    })
    .background_gradient(
        subset=["Matched degradation (subject)"],

```

```

        cmap="RdYlBu_r",
        vmin=-0.15,
        vmax=0.40,
    )
    .background_gradient(
        subset=["Subject F1"],
        cmap="Blues",
        vmin=0.45,
        vmax=1.0,
    )
    .set_caption("Cross-condition transfer summary (subject-primary columns_
    emphasized)")
)

display_existing_figure("step3", "degradation_heatmap", width=780)
display_existing_figure("step3", "cross_condition_f1_heatmap", width=780)
display_existing_figure("step3", "step3_subject_confusion_matrices", width=780)
display_existing_figure("step3", "step3_direction_intervals", width=780)
display_existing_figure("step3", "step3_bidirectional_asymmetry", width=780)
display(styled_transfer)

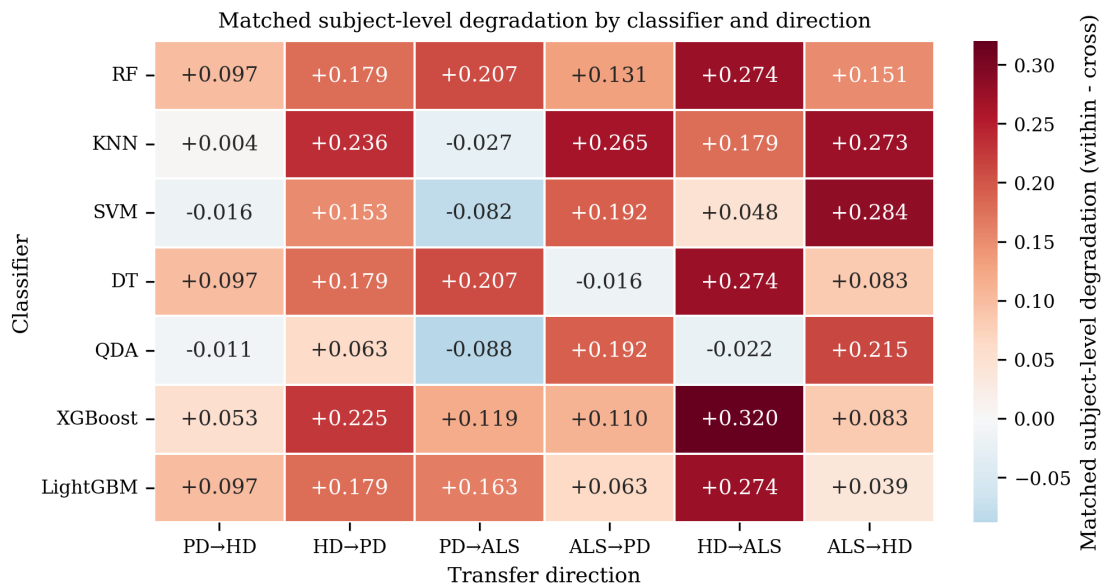
```

Saved table: report/tables/v4/paper_main_results_summary_v4.csv

Saved table: report/tables/v4/paper_main_results_summary_v4.tex

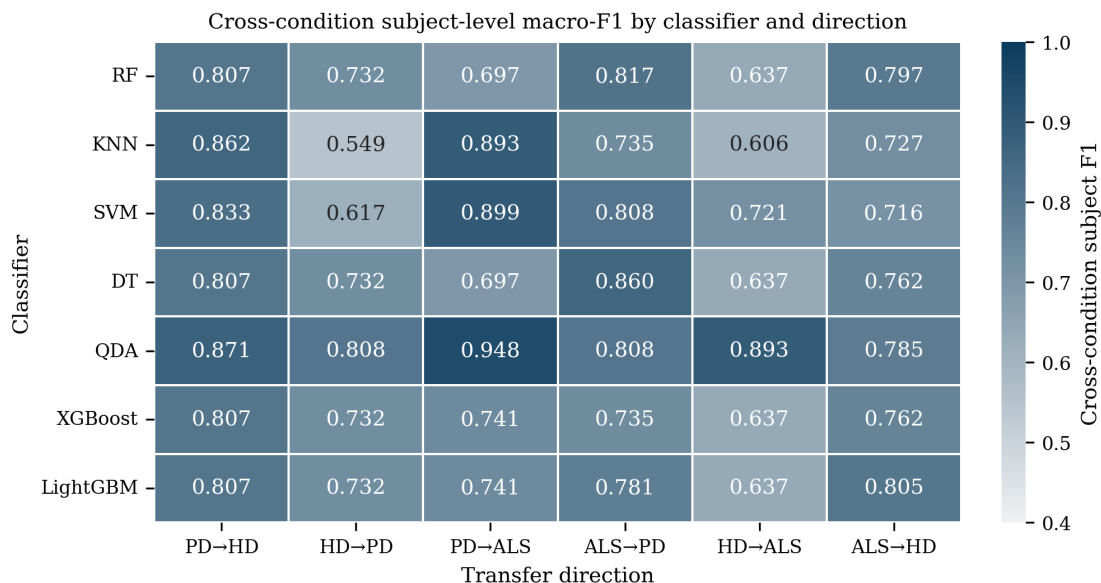
Displayed figure: report/figures/v4/png/step3/degradation_heatmap.png

Vector source: report/figures/v4/pdf/step3/degradation_heatmap.pdf



Displayed figure: report/figures/v4/png/step3/cross_condition_f1_heatmap.png

Vector source: report/figures/v4/pdf/step3/cross_condition_f1_heatmap.pdf

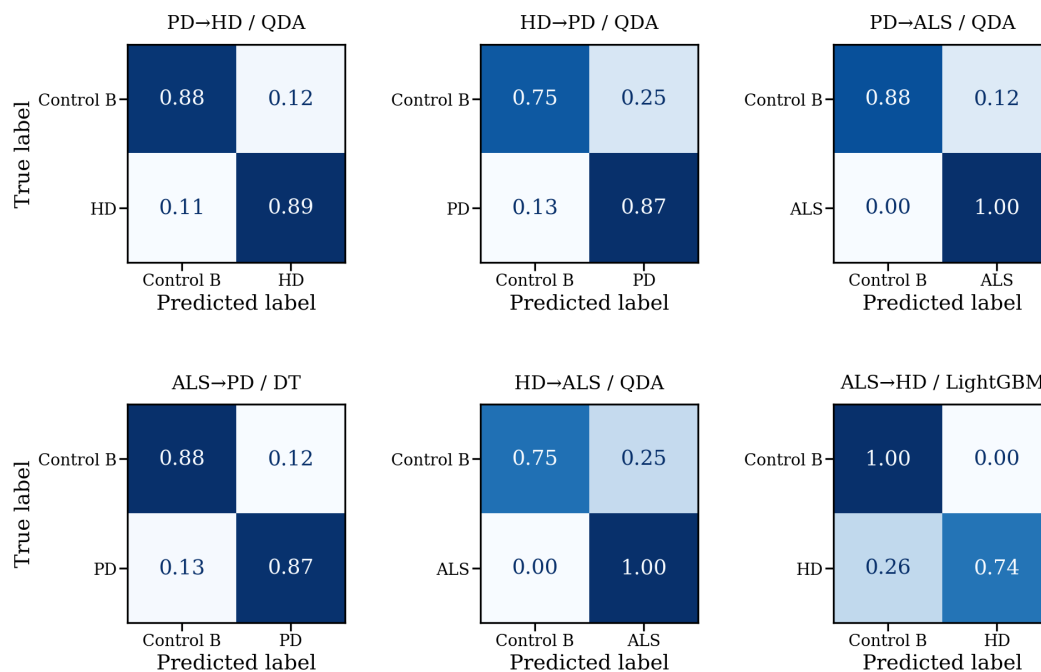


Displayed figure:

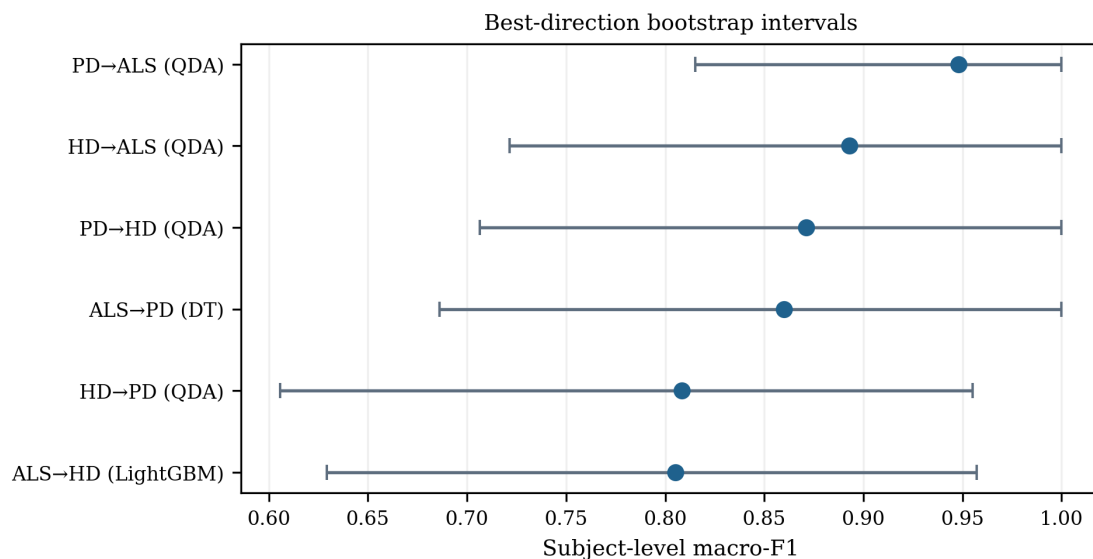
`report/figures/v4/png/step3/step3_subject_confusion_matrices.png`

Vector source: `report/figures/v4/pdf/step3/step3_subject_confusion_matrices.pdf`

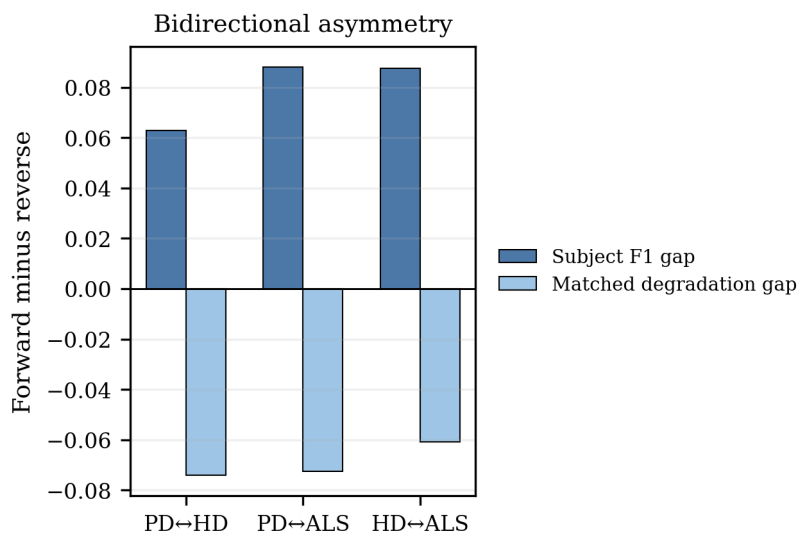
Subject-level normalized confusion matrices for the best classifier in each direction



Displayed figure: report/figures/v4/png/step3/step3_direction_intervals.png
 Vector source: report/figures/v4/pdf/step3/step3_direction_intervals.pdf



Displayed figure: report/figures/v4/png/step3/step3_bidirectional_asymmetry.png
 Vector source: report/figures/v4/pdf/step3/step3_bidirectional_asymmetry.pdf



<pandas.io.formats.style.Styler at 0x7f6e32f5f680>

The transfer figures support the central v4 claim directly. At the direction-average level, **matched subject-level degradation is positive in all six directions**, with the largest mean losses in **HD→ALS (0.1925)** and **HD→PD (0.1735)**, and the smallest in **PD→HD (0.0458)**. The ab-

solute subject-F1 heatmap shows that transfer remains practically strong in selected directions even when degradation is positive overall; the best direction-level outcomes are **QDA** for **PD→ALS** (0.9481), **HD→ALS** (0.8929), **PD→HD** (0.8712), and **HD→PD** (0.8083), with **DT** leading **ALS→PD** (0.8600) and **LightGBM** leading **ALS→HD** (0.8052). The confusion matrices add a complementary class-balance view: for example, **ALS→HD** achieves perfect control specificity at the cost of lower HD recall, whereas **HD→ALS** reaches full ALS recall with lower control specificity. The key caveat remains explicit in the table: some individual classifier-direction rows have near-zero or negative matched degradation even though the direction-level means are uniformly positive.

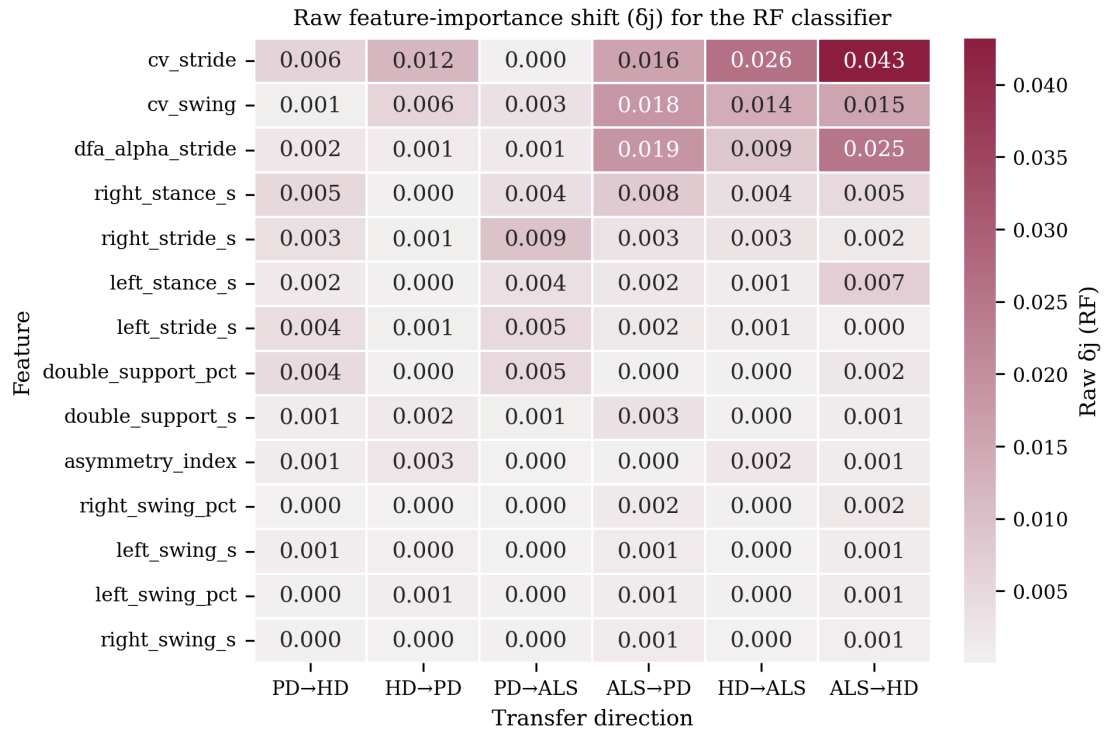
1.5 Step 4: SHAP-based transfer diagnostics

Step 4 should stay diagnostic rather than causal. The manuscript therefore benefits from showing the model-specific random-forest map, the cross-classifier consensus map, and the rank-agreement panel together, with family-level and recurrence summaries available as supporting detail.

```
[28]: top_features = pd.read_csv(TABLE_DIR / "step4_top_feature_recurrence_v4.csv")
family_delta = pd.read_csv(TABLE_DIR / "step4_family_delta_j_long_v4.csv")
family_means = (
    family_delta.groupby("family")["delta_j_family"]
    .mean()
    .sort_values(ascending=False)
    .round(4)
    .rename("mean_delta_j_family")
    .reset_index()
)

display_existing_figure("step4", "step4_delta_j_rf_heatmap", width=780)
display_existing_figure("step4", "step4_delta_j_normalized_consensus_heatmap",
    ↪width=780)
display_existing_figure("step4", "step4_delta_j_spearman", width=780)
display_existing_figure("step4", "step4_family_delta_j_consensus", width=780)
display_existing_figure("step4", "step4_top_feature_recurrence", width=780)
display(top_features.head(10))
display(family_means)
```

Displayed figure: report/figures/v4/png/step4/step4_delta_j_rf_heatmap.png
 Vector source: report/figures/v4/pdf/step4/step4_delta_j_rf_heatmap.pdf

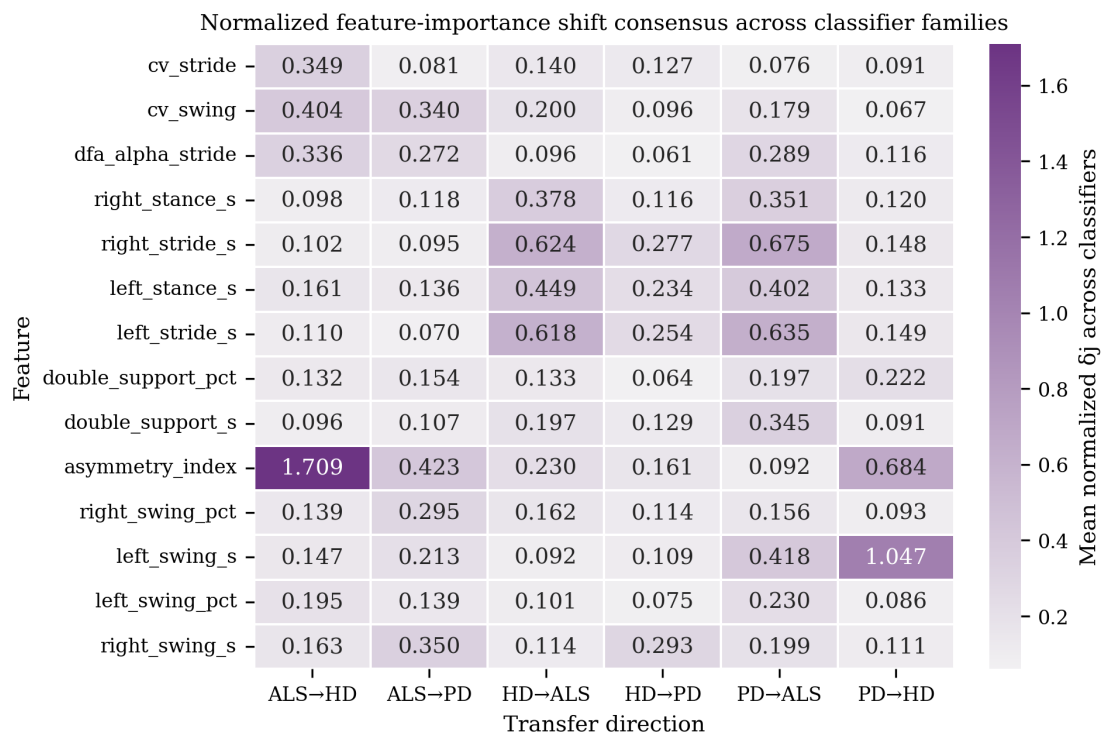


Displayed figure:

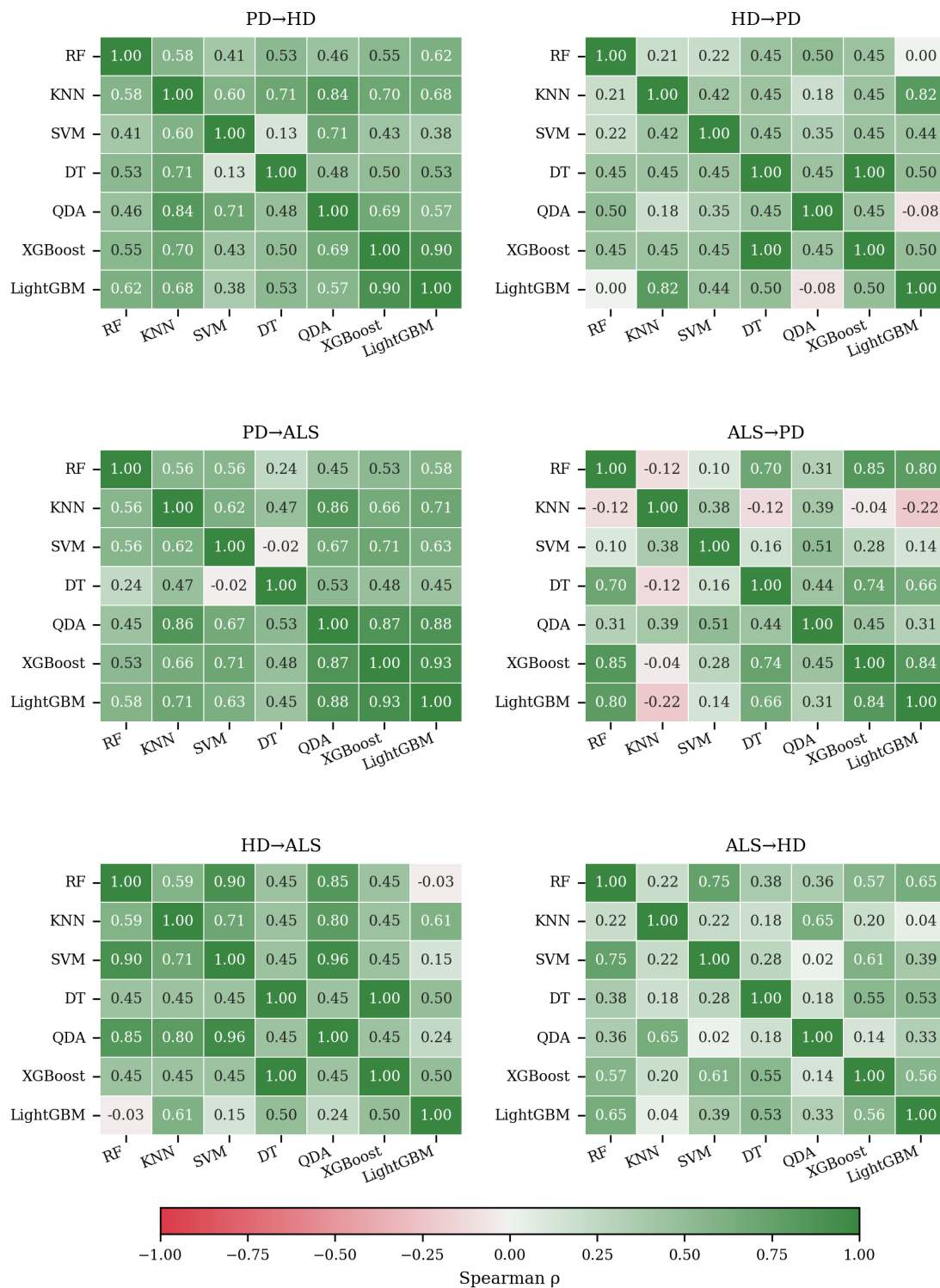
report/figures/v4/png/step4/step4_delta_j_normalized_consensus_heatmap.png

Vector source:

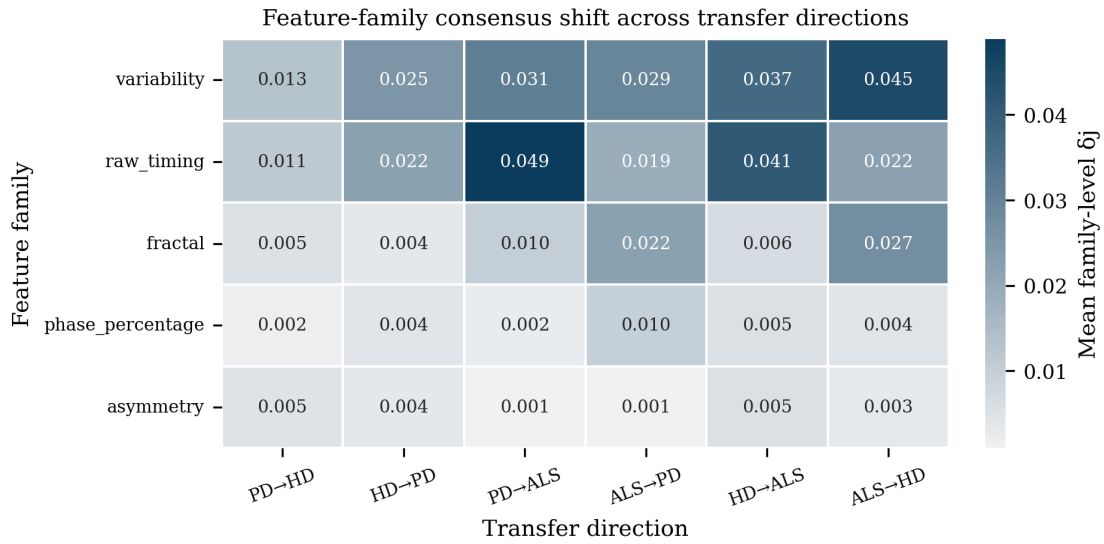
report/figures/v4/pdf/step4/step4_delta_j_normalized_consensus_heatmap.pdf



Displayed figure: report/figures/v4/png/step4/step4_delta_j_spearman.png
 Vector source: report/figures/v4/pdf/step4/step4_delta_j_spearman.pdf

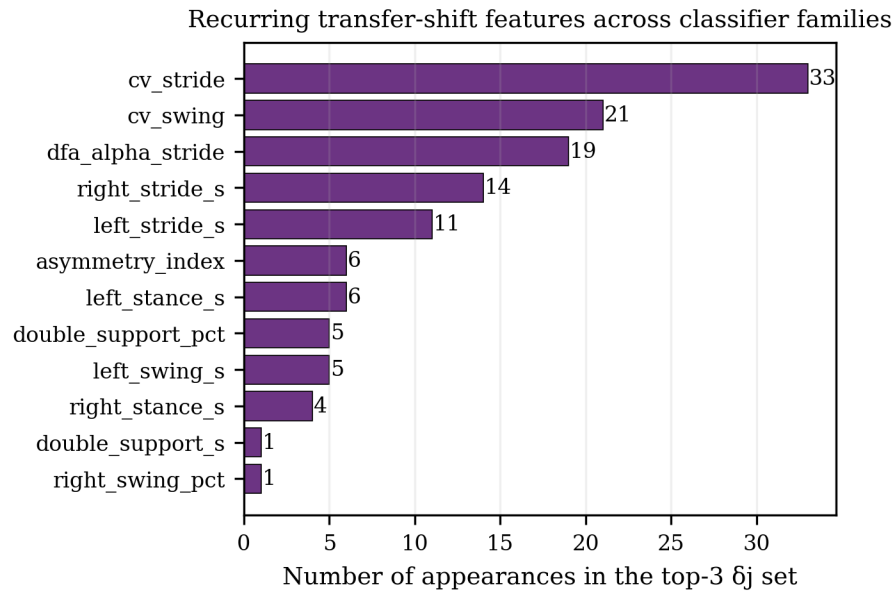


Displayed figure: report/figures/v4/png/step4/step4_family_delta_j_consensus.png
 Vector source: report/figures/v4/pdf/step4/step4_family_delta_j_consensus.pdf



Displayed figure: report/figures/v4/png/step4/step4_top_feature_recurrence.png

Vector source: report/figures/v4/pdf/step4/step4_top_feature_recurrence.pdf



	feature	top3_count
0	cv_stride	33
1	cv_swing	21
2	dfa_alpha_stride	19
3	right_stride_s	14
4	left_stride_s	11

5	asymmetry_index	6
6	left_stance_s	6
7	double_support_pct	5
8	left_swing_s	5
9	right_stance_s	4
	family	mean_delta_j_family
0	variability	0.0302
1	raw_timing	0.0274
2	fractal	0.0124
3	phase_percentage	0.0047
4	asymmetry	0.0031

The Step 4 diagnostic layer is coherent but should still be described conservatively. The consensus heatmap and recurrence table point repeatedly to **cv_stride** (33 top-3 appearances), **cv_swing** (21), and **dfa_alpha_stride** (19) as recurring transfer-shift features. At the family level, the mean movement is largest for **variability** (0.0302) and **raw timing** (0.0274), with **fractal** movement still visible (0.0124) and the remaining families smaller. The Spearman panel is useful because it distinguishes directions where classifier families tell a similar diagnostic story from directions where the feature rankings are more model contingent. The important caveat is unchanged: these figures diagnose changes in feature reliance under transfer, not disease-specific causal biomarkers.

1.6 Step 5: stress testing, feature sensitivity, and exploratory conformal diagnostics

Step 5 covers several genuinely different questions, so the synthesis notebook keeps more than one main figure here. The selected panels separate three claims: how transfer degrades under Gaussian feature-space stress, which individual features drive strong permutation sensitivity, and how exploratory conformal set behavior changes between matched and shifted evaluation.

```
[29]: noise_summary = pd.read_csv(TABLE_DIR / "step5_noise_robustness_summary_v4.csv")
family_sensitivity = pd.read_csv(TABLE_DIR / "step5_feature_family_sensitivity_v4.csv")
conformal_summary = pd.read_csv(TABLE_DIR / "step5_conformal_summary_v4.csv")
conformal_methods = conformal_summary[conformal_summary["method"].isin(["aps", "lac"])]
conformal_agg = (
    conformal_methods
    .groupby(["scope", "method", "alpha"])[
        [
            "coverage_subject_level",
            "raw_mean_set_size",
            "post_mean_set_size",
            "raw_empty_rate",
            "post_empty_rate",
        ]
    ]
    .mean()
    .reset_index()
```

```

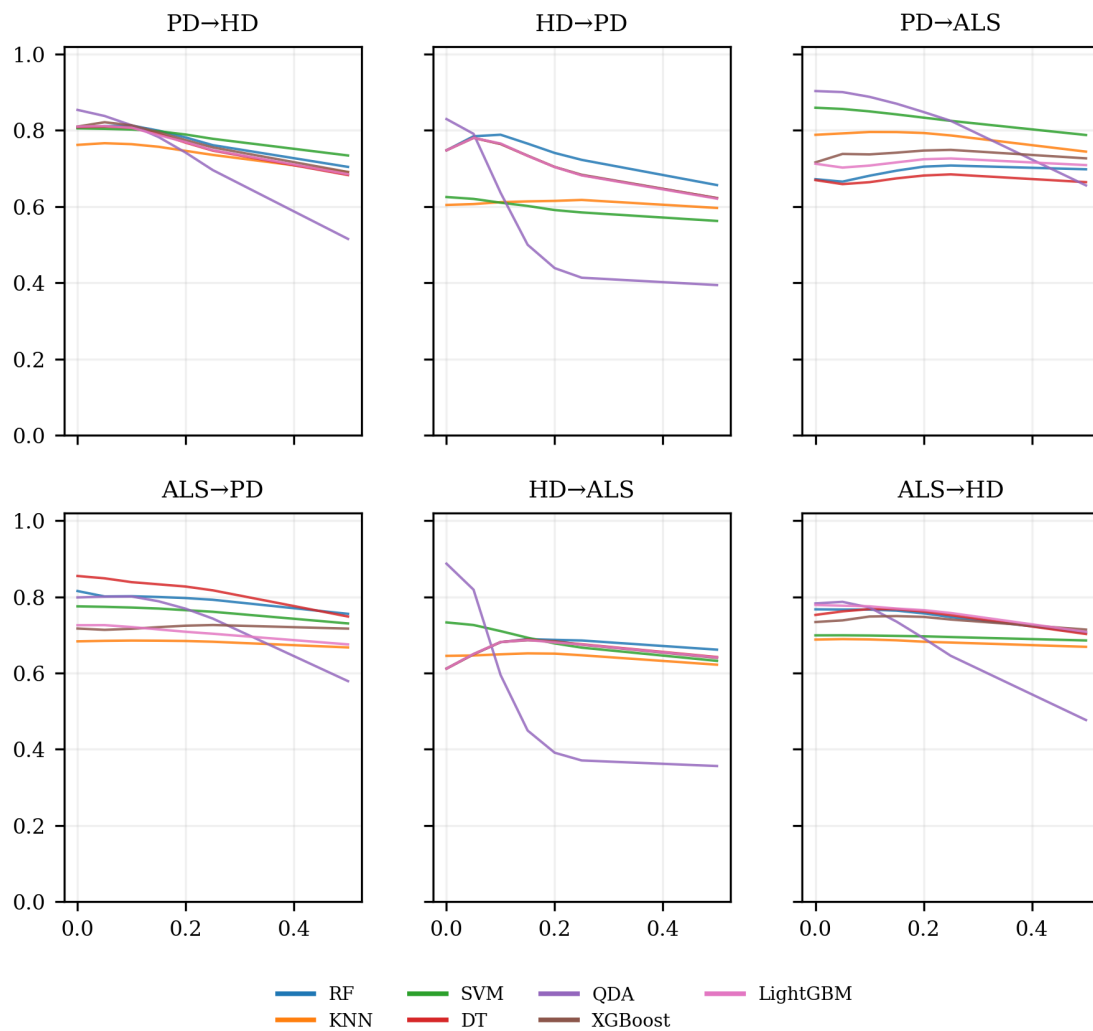
        .round(4)
    )
    family_drop_means = (
        family_sensitivity.groupby(["scope", "family"])["drop"]
        .mean()
        .reset_index()
        .sort_values(["scope", "drop"], ascending=[True, False])
        .round(4)
    )

    display_existing_figure("step5", "noise_curves_cross", width=780)
    display_existing_figure("step5", "step5_feature_sensitivity_within", width=780)
    display_existing_figure("step5", "step5_feature_sensitivity_cross", width=780)
    display_existing_figure("step5", "conformal_within", width=780)
    display_existing_figure("step5", "conformal_cross", width=780)
    display_existing_figure("step5", "step5_feature_family_sensitivity", width=780)
    display_existing_figure("step5", "step5_corruption_cross", width=780)
    display(noise_summary.head(12))
    display(family_drop_means)
    display(conformal_agg)

```

Displayed figure: report/figures/v4/png/step5/noise_curves_cross.png

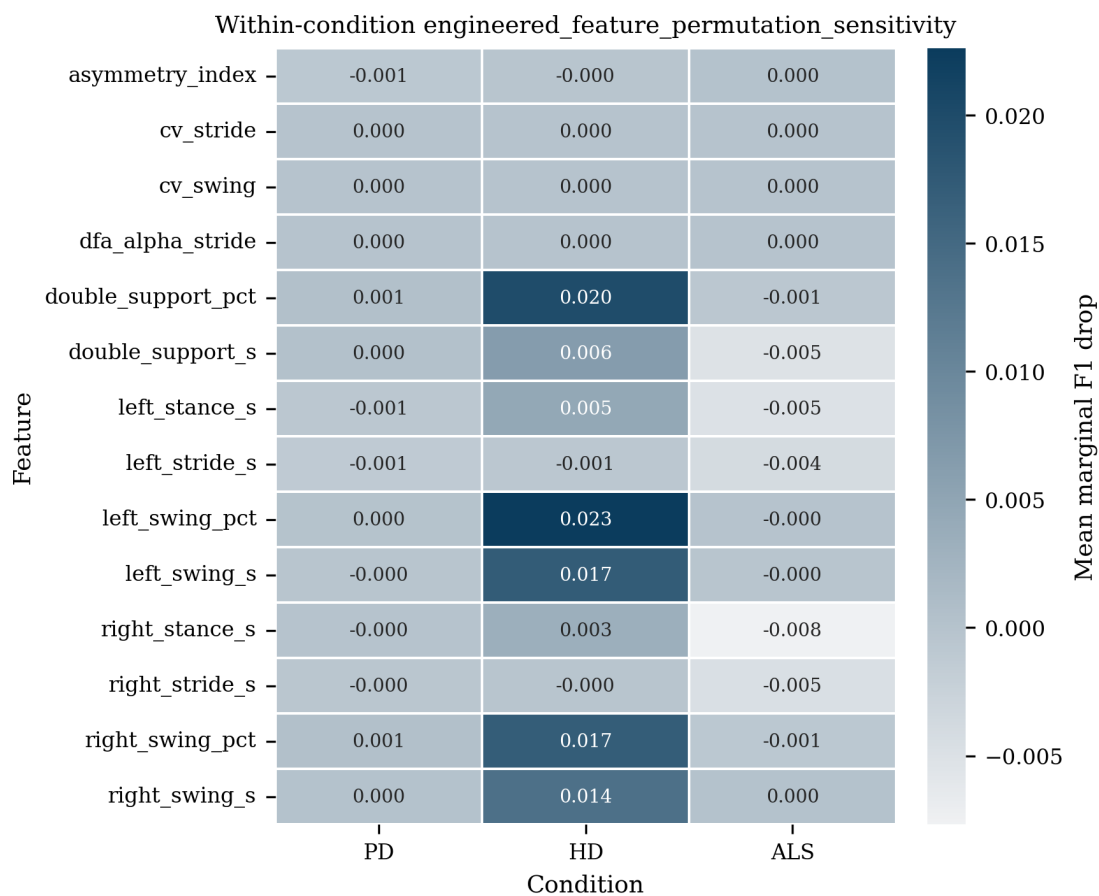
Vector source: report/figures/v4/pdf/step5/noise_curves_cross.pdf



Displayed figure:

report/figures/v4/png/step5/step5_feature_sensitivity_within.png

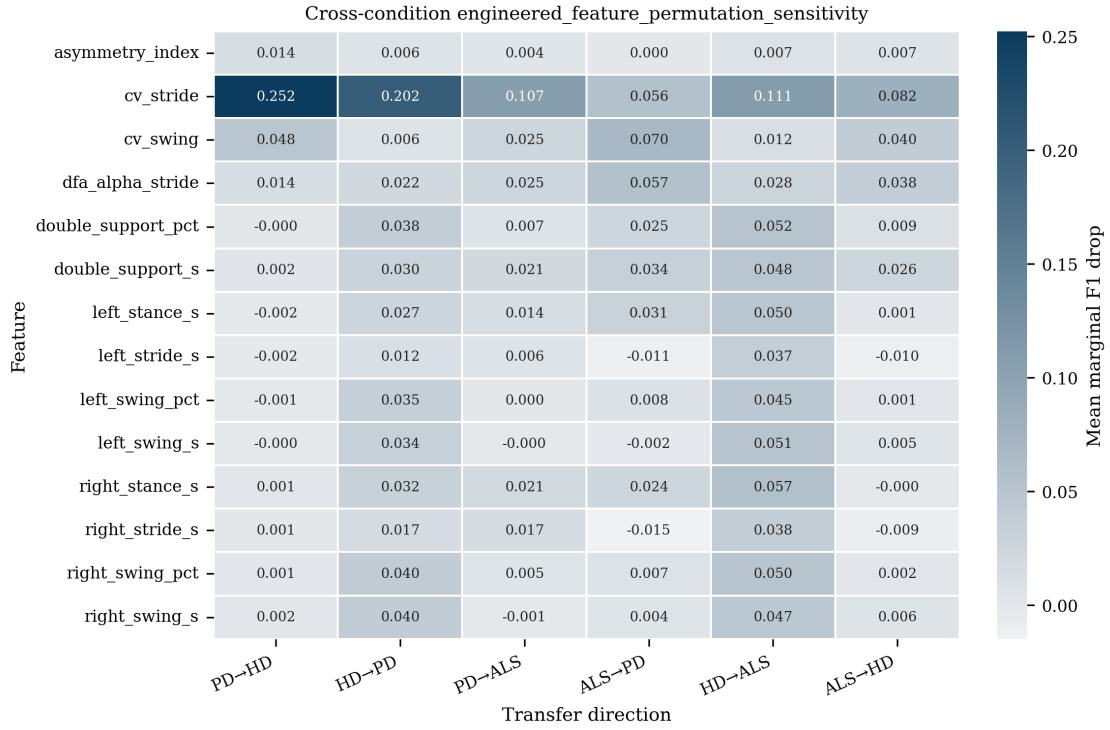
Vector source: report/figures/v4/pdf/step5/step5_feature_sensitivity_within.pdf



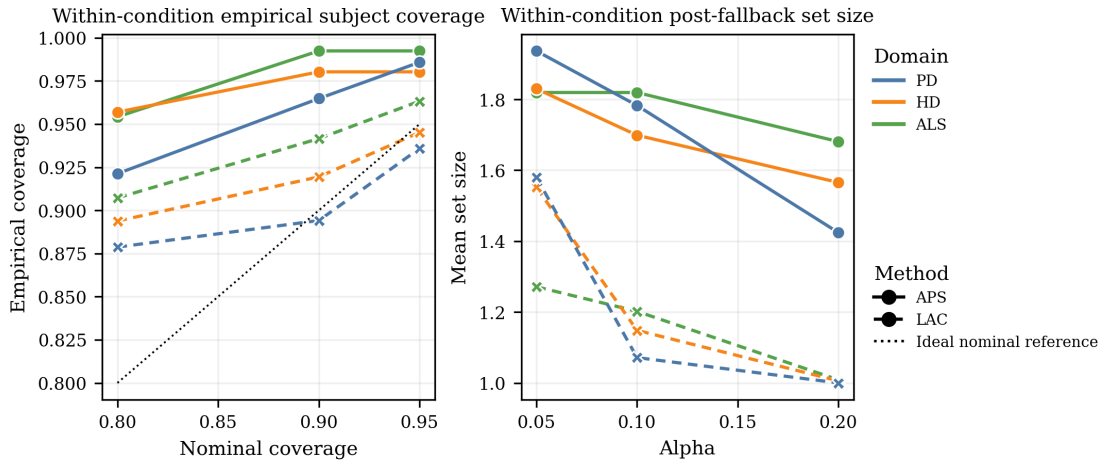
Displayed figure:

report/figures/v4/png/step5/step5_feature_sensitivity_cross.png

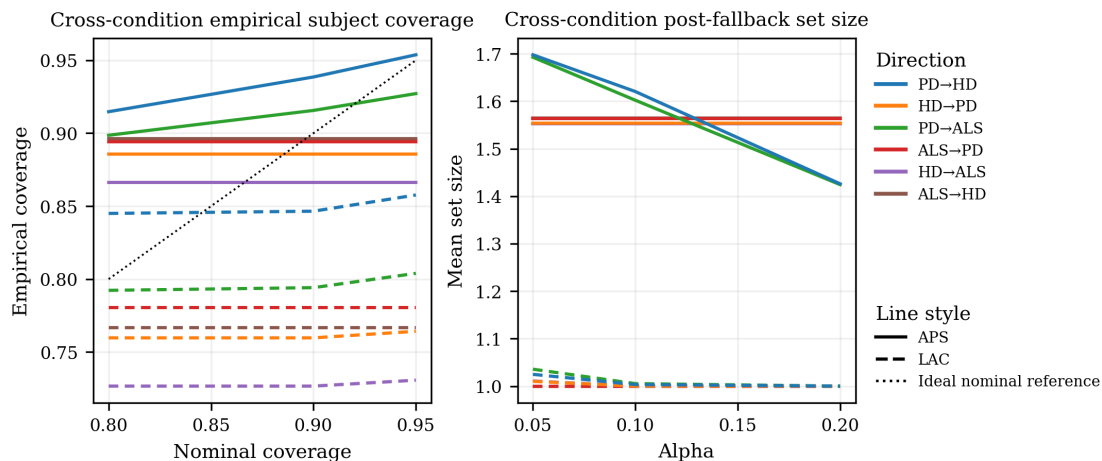
Vector source: report/figures/v4/pdf/step5/step5_feature_sensitivity_cross.pdf



Displayed figure: report/figures/v4/png/step5/conformal_within.png
 Vector source: report/figures/v4/pdf/step5/conformal_within.pdf



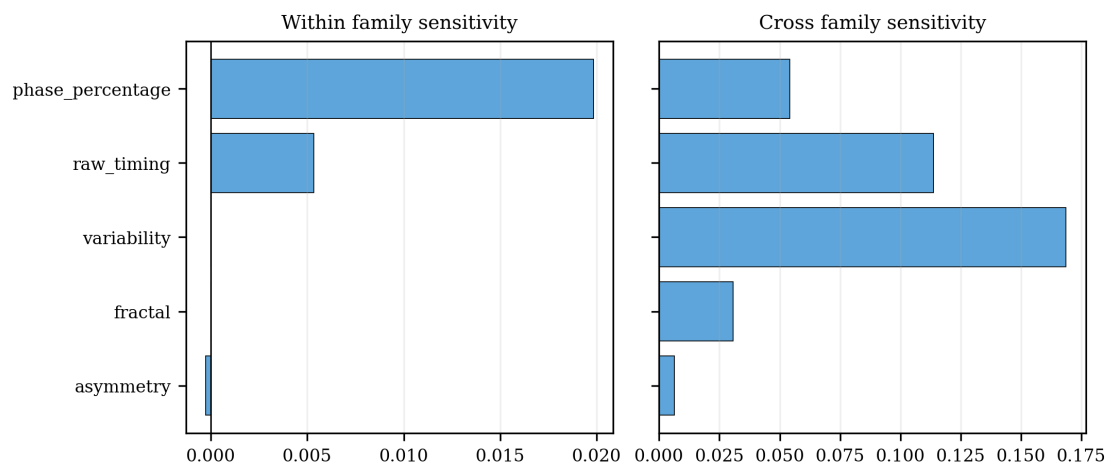
Displayed figure: report/figures/v4/png/step5/conformal_cross.png
 Vector source: report/figures/v4/pdf/step5/conformal_cross.pdf



Displayed figure:

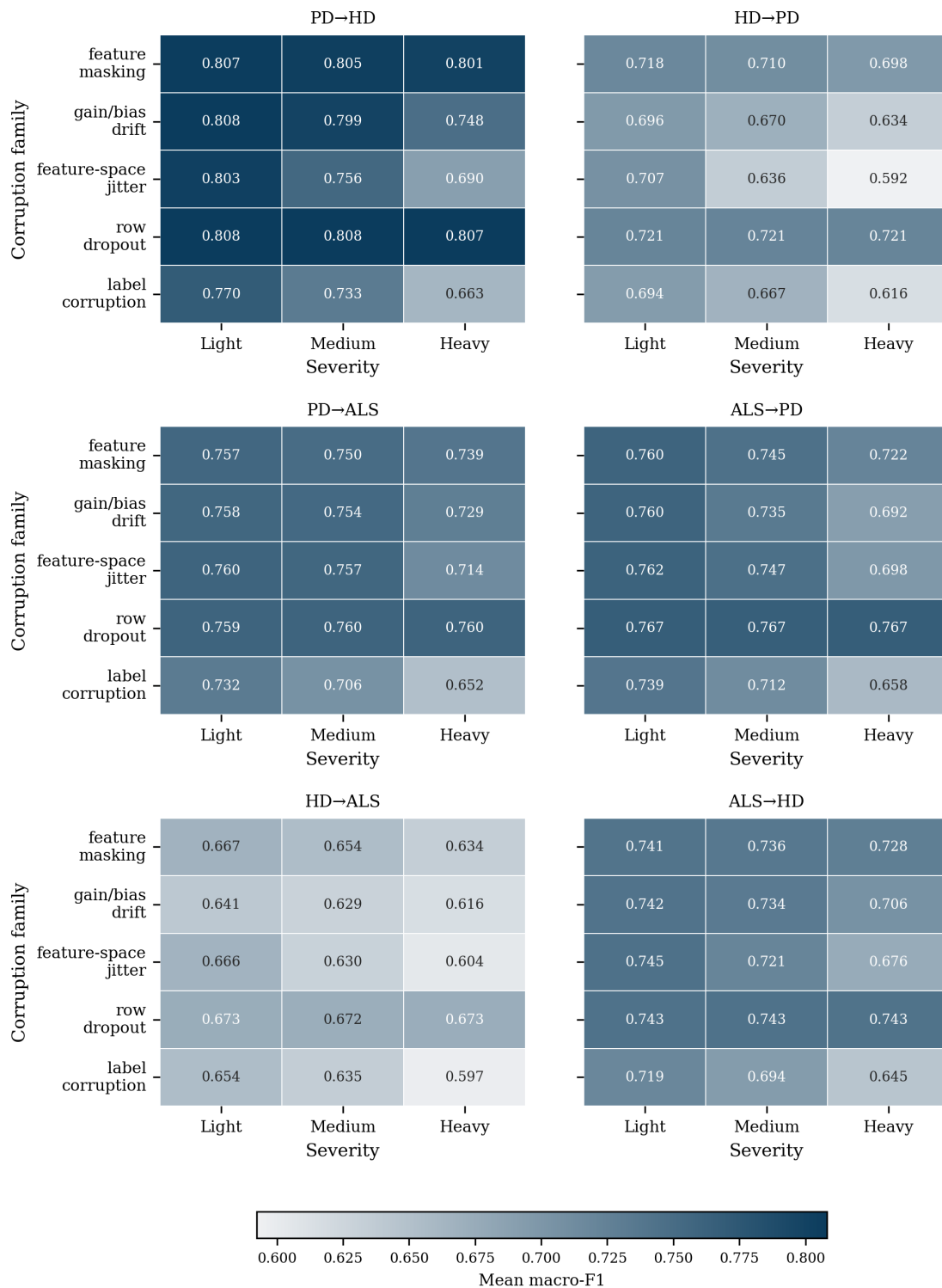
report/figures/v4/png/step5/step5_feature_family_sensitivity.png

Vector source: report/figures/v4/pdf/step5/step5_feature_family_sensitivity.pdf



Displayed figure: report/figures/v4/png/step5/step5_corruption_cross.png

Vector source: report/figures/v4/pdf/step5/step5_corruption_cross.pdf



```

scope classifier      f100  f100.10  f100.25  f100.50  \
0  within:pd         rf  0.899398  0.875041  0.823609  0.748199

```

1	within:pd	knn	0.875902	0.871854	0.853874	0.795882
2	within:pd	svm	0.843980	0.836010	0.819342	0.777173
3	within:pd	dt	0.902162	0.881461	0.802473	0.714832
4	within:pd	qda	0.835888	0.841176	0.829259	0.742713
5	within:pd	xgb	0.850855	0.856269	0.797043	0.726743
6	within:pd	lgbm	0.901946	0.884440	0.812255	0.733684
7	within:hd	rf	0.904350	0.868198	0.792960	0.727331
8	within:hd	knn	0.752501	0.743426	0.713033	0.678451
9	within:hd	svm	0.854348	0.811300	0.735511	0.679115
10	within:hd	dt	0.904350	0.865778	0.782523	0.709218
11	within:hd	qda	0.873218	0.621495	0.417904	0.408133

	delta_f1@0.25	sigma_at_10pct_drop
0	0.075789	0.50
1	0.022028	NaN
2	0.024638	NaN
3	0.099689	0.25
4	0.006629	0.50
5	0.053812	0.50
6	0.089691	0.50
7	0.111390	0.20
8	0.039468	NaN
9	0.118837	0.20
10	0.121827	0.20
11	0.455314	0.10

	scope	family	drop
4	cross	variability	0.1685
3	cross	raw_timing	0.1137
2	cross	phase_percentage	0.0541
1	cross	fractal	0.0307
0	cross	asymmetry	0.0063
7	within	phase_percentage	0.0198
8	within	raw_timing	0.0053
6	within	fractal	0.0000
9	within	variability	0.0000
5	within	asymmetry	-0.0003

	scope	method	alpha	coverage_subject_level	raw_mean_set_size \
0	cross	aps	0.05	0.9038	1.5985
1	cross	aps	0.10	0.8994	1.5667
2	cross	aps	0.20	0.8926	1.4516
3	cross	lac	0.05	0.7840	0.7789
4	cross	lac	0.10	0.7791	0.7163
5	cross	lac	0.20	0.7785	0.6560
6	within	aps	0.05	0.9862	1.8546
7	within	aps	0.10	0.9791	1.7492
8	within	aps	0.20	0.9441	1.5152
9	within	lac	0.05	0.9481	1.4622

10	within	lac	0.10	0.9184	1.1146
11	within	lac	0.20	0.8932	0.9027

	post_mean_set_size	raw_empty_rate	post_empty_rate
0	1.6043	0.0057	0.0
1	1.5763	0.0095	0.0
2	1.5142	0.0626	0.0
3	1.0135	0.2347	0.0
4	1.0014	0.2851	0.0
5	1.0000	0.3440	0.0
6	1.8620	0.0073	0.0
7	1.7669	0.0177	0.0
8	1.5567	0.0415	0.0
9	1.4680	0.0058	0.0
10	1.1405	0.0259	0.0
11	1.0048	0.1021	0.0

The Step 5 evidence is rich enough that several panels deserve main-paper status. The cross-condition noise curves show that transferred performance is stress-sensitive in a direction-dependent way rather than collapsing uniformly. The feature-sensitivity heatmaps and family summaries sharpen that story: under transfer, the largest average family-level drops are in **variability (0.1685)** and **raw timing (0.1137)**, whereas within-condition family effects are generally smaller and more selective. The exploratory conformal panels add an uncertainty-behavior layer. Averaged across domains, **APS** retains larger mean set sizes and higher subject-level coverage than **LAC** both within-condition and under transfer, while **LAC** becomes notably tighter and more empty-set-prone before fallback under shift (raw empty rates rising to about **0.34** at $\alpha=0.20$). The final caveat is crucial: this entire step is diagnostic stress testing. Gaussian feature-space perturbation is not raw sensor noise, benchmark label corruption is not noisy-label training, and cross-condition conformal behavior is exploratory under distribution shift.

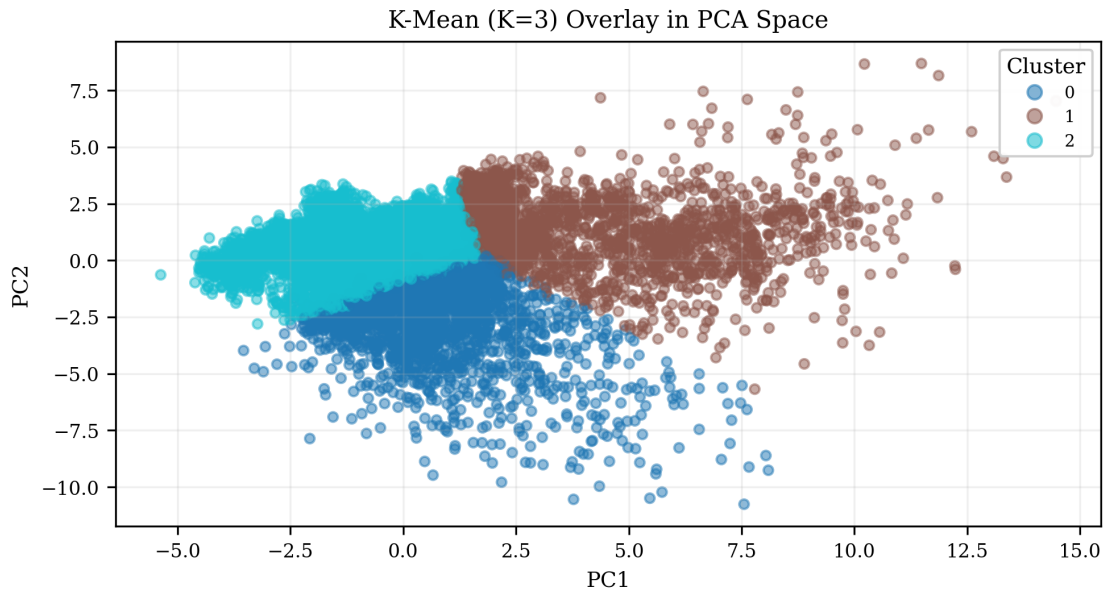
1.7 Step 6: exploratory geometry

Step 6 remains exploratory, but one geometry panel is still worth surfacing in the synthesis notebook because it provides qualitative context for the subject-level feature space without claiming supervised evidence.

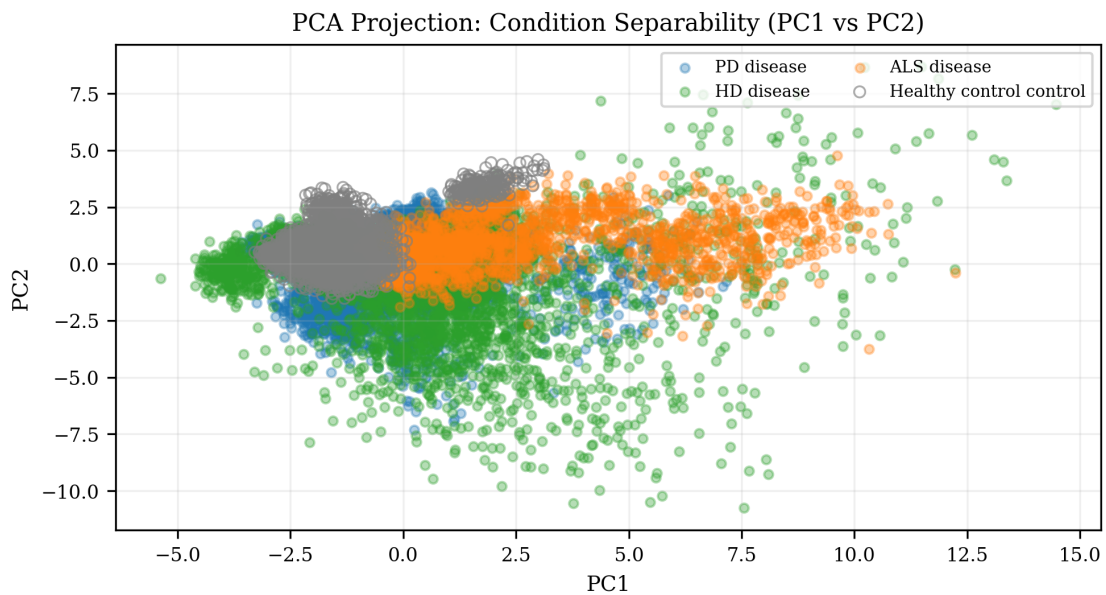
```
[30]: step6_summary = json.loads((REPO_ROOT / "experiments" / "results" / "v4" /
↳ "step6_subject_views_summary.json").read_text())
display_existing_figure("step6", "kmeans_scatter_k3", width=780)
display_existing_figure("step6", "pca_scatter_by_condition", width=780)
display(pd.DataFrame(step6_summary["views"]).T[[
    "subject_count",
    "ari_against_relevant_labels",
    "seed_stability_mean_ari",
    "subject_bootstrap_stability_mean_ari",
]])
```

Displayed figure: report/figures/v4/png/step6/kmeans_scatter_k3.png

Vector source: report/figures/v4/pdf/step6/kmeans_scatter_k3.pdf



Displayed figure: report/figures/v4/png/step6/pca_scatter_by_condition.png
 Vector source: report/figures/v4/pdf/step6/pca_scatter_by_condition.pdf



	subject_count	ari_against_relevant_labels	\
disease_only	47	0.013473	
all_subjects	63	0.025964	
binary_pathological_vs_control	63	-0.099877	

control_a_vs_control_b	16	0.0
------------------------	----	-----

	seed_stability_mean_ari	\
disease_only	0.95945	
all_subjects	0.982461	
binary_pathological_vs_control	0.982461	
control_a_vs_control_b	1.0	

	subject_bootstrap_stability_mean_ari
disease_only	0.058784
all_subjects	0.043274
binary_pathological_vs_control	-0.067291
control_a_vs_control_b	0.073609

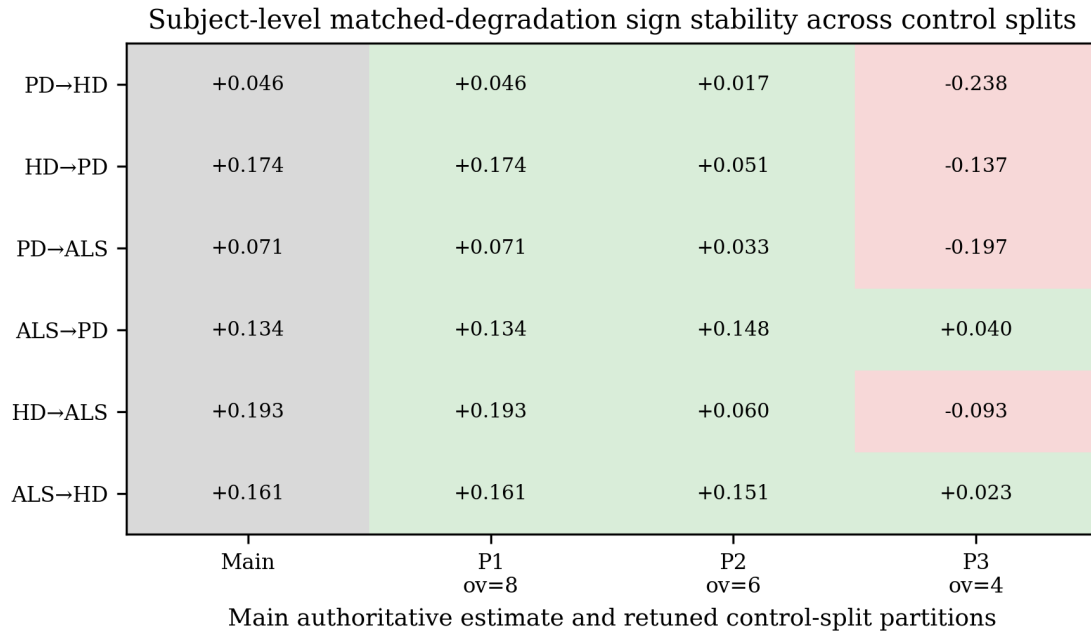
The exploratory geometry panels are visually informative but scientifically limited. The **K-means overlay in PCA space** is the strongest Step 6 figure for the manuscript because it shows that the subject-level feature space has visible structure, yet the saved summary metrics keep the interpretation honest: ARI against the relevant labels remains weak (**0.0260** for `all_subjects`, **0.0135** for `disease_only`), and bootstrap stability is much lower than seed-to-seed stability. In other words, the geometry is useful context for transfer difficulty, but it does not prove disease separability or replace the supervised transfer results.

1.8 Step 8: control-partition sensitivity

Step 8 is now a required part of the final paper story because it tests whether the main transfer pattern is stable when the control partition changes within the near-optimal candidate family. The key framing is sensitivity, not invariance.

```
[31]: step8_sensitivity = pd.read_csv(TABLE_DIR / "step8_direction_sensitivity_v4.
      ↪csv")
step8_within = pd.read_csv(TABLE_DIR / "step8_within_baselines_v4.csv")
display_existing_figure("step8", "step8_sign_stability_matrix", width=780)
display_existing_figure("step8", "step8_direction_degradation_ranges", ↪
      ↪width=780)
display_existing_figure("step8", "step8_within_source_baselines", width=780)
display_existing_figure("step8", "step8_best_transfer_leaders_heatmap", ↪
      ↪width=780)
display(step8_sensitivity)
display(step8_within)
```

Displayed figure: report/figures/v4/png/step8/step8_sign_stability_matrix.png
 Vector source: report/figures/v4/pdf/step8/step8_sign_stability_matrix.pdf

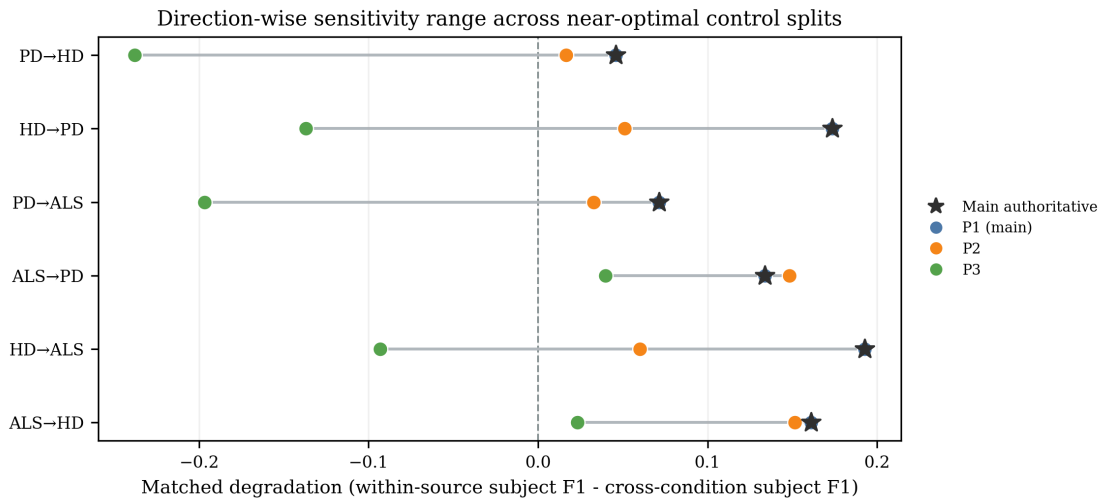


Displayed figure:

report/figures/v4/png/step8/step8_direction_degradation_ranges.png

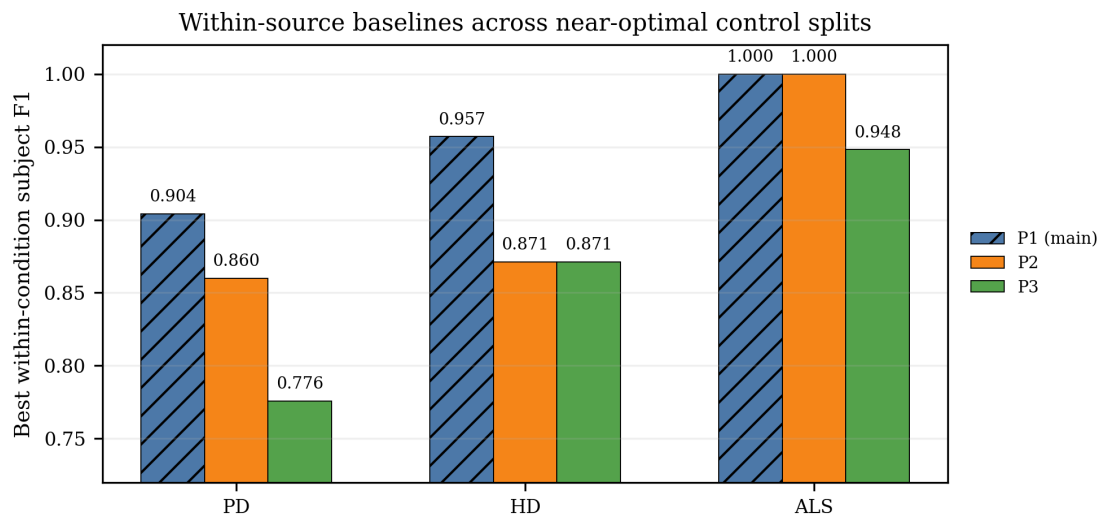
Vector source:

report/figures/v4/pdf/step8/step8_direction_degradation_ranges.pdf



Displayed figure: report/figures/v4/png/step8/step8_within_source_baselines.png

Vector source: report/figures/v4/pdf/step8/step8_within_source_baselines.pdf

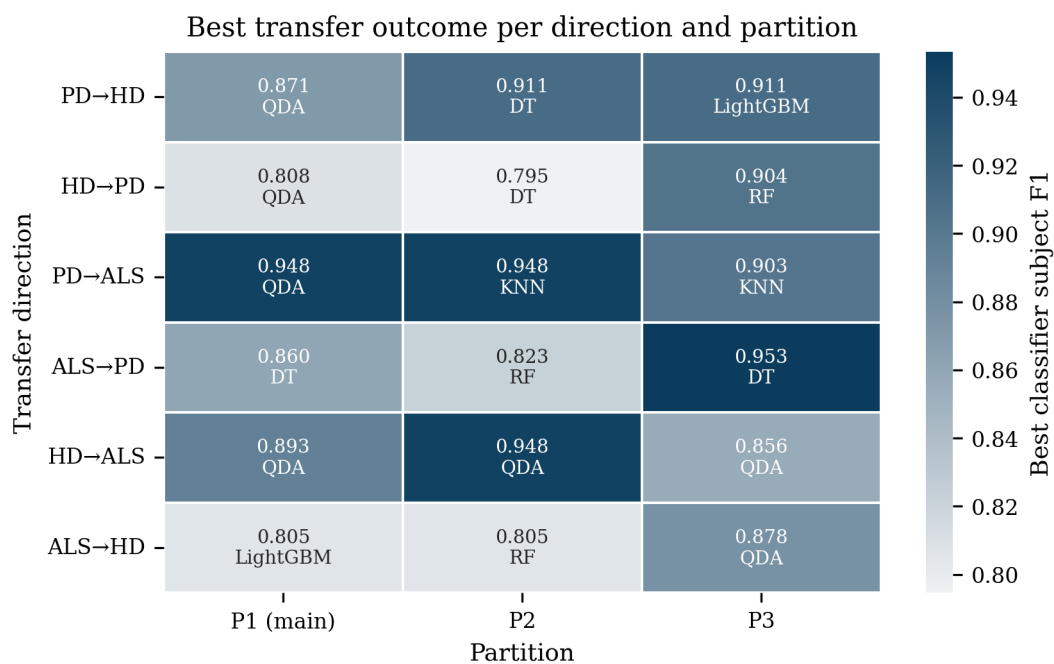


Displayed figure:

report/figures/v4/png/step8/step8_best_transfer_leaders_heatmap.png

Vector source:

report/figures/v4/pdf/step8/step8_best_transfer_leaders_heatmap.pdf



	partition_index	partition_label	direction	direction_label	\
0	1	P1 (main)	pd_to_hd	PD→HD	
1	1	P1 (main)	hd_to_pd	HD→PD	

2	1	P1 (main)	pd_to_als	PD→ALS
3	1	P1 (main)	als_to_pd	ALS→PD
4	1	P1 (main)	hd_to_als	HD→ALS
5	1	P1 (main)	als_to_hd	ALS→HD
6	2	P2	pd_to_hd	PD→HD
7	2	P2	hd_to_pd	HD→PD
8	2	P2	pd_to_als	PD→ALS
9	2	P2	als_to_pd	ALS→PD
10	2	P2	hd_to_als	HD→ALS
11	2	P2	als_to_hd	ALS→HD
12	3	P3	pd_to_hd	PD→HD
13	3	P3	hd_to_pd	HD→PD
14	3	P3	pd_to_als	PD→ALS
15	3	P3	als_to_pd	ALS→PD
16	3	P3	hd_to_als	HD→ALS
17	3	P3	als_to_hd	ALS→HD

	mean_matched_degradation_subject	median_matched_degradation_subject	\
0	0.045839	0.052898	
1	0.173528	0.179249	
2	0.071398	0.119300	
3	0.133876	0.130688	
4	0.192531	0.274147	
5	0.161047	0.151156	
6	0.016548	0.020468	
7	0.051010	0.055634	
8	0.032853	0.046057	
9	0.148368	0.182540	
10	0.060071	0.136218	
11	0.151488	0.142953	
12	-0.237821	-0.246515	
13	-0.137059	-0.162930	
14	-0.196728	-0.198102	
15	0.039600	0.082651	
16	-0.093248	-0.107538	
17	0.023142	0.039249	

	main_authoritative_subject_delta	deviation_from_main	direction_sign	\
0	0.045839	0.000000	+	
1	0.173528	0.000000	+	
2	0.071398	0.000000	+	
3	0.133876	0.000000	+	
4	0.192531	0.000000	+	
5	0.161047	0.000000	+	
6	0.045839	-0.029291	+	
7	0.173528	-0.122518	+	
8	0.071398	-0.038545	+	
9	0.133876	0.014492	+	

10	0.192531	-0.132460	+
11	0.161047	-0.009559	+
12	0.045839	-0.283660	-
13	0.173528	-0.310587	-
14	0.071398	-0.268126	-
15	0.133876	-0.094276	+
16	0.192531	-0.285779	-
17	0.161047	-0.137905	+

	best_classifier	best_classifier_label	best_subject_f1	best_delta_subject
0	qda	QDA	0.871224	-0.011183
1	qda	QDA	0.808333	0.062891
2	qda	QDA	0.948148	-0.088107
3	dt	DT	0.860041	-0.015597
4	qda	QDA	0.892857	-0.021633
5	lgbm	LightGBM	0.805195	0.039249
6	dt	DT	0.911184	-0.059571
7	dt	DT	0.794643	0.076581
8	knn	KNN	0.948148	-0.130688
9	rf	RF	0.823077	0.125071
10	qda	QDA	0.948148	-0.151156
11	rf	RF	0.805195	0.142953
12	lgbm	LightGBM	0.911184	-0.356345
13	rf	RF	0.904167	-0.258764
14	knn	KNN	0.902778	-0.190278
15	dt	DT	0.953347	-0.193622
16	qda	QDA	0.855835	-0.189168
17	qda	QDA	0.878195	-0.137454

	partition_index	partition_label	condition	condition_label	best_subject_f1
0	1	P1 (main)	pd	PD	0.904167
1	1	P1 (main)	hd	HD	0.957075
2	1	P1 (main)	als	ALS	1.000000
3	2	P2	pd	PD	0.860041
4	2	P2	hd	HD	0.871224
5	2	P2	als	ALS	1.000000
6	3	P3	pd	PD	0.775828
7	3	P3	hd	HD	0.871224
8	3	P3	als	ALS	0.948148

The Step 8 result is scientifically important precisely because it is mixed. **Partition 2 preserves the direction-level matched-degradation signs**, but **partition 3 reverses four of six signs**, so the control-partition sensitivity analysis clearly **qualifies** the main transfer story rather than proving invariance. The within-source baseline panel explains part of that sensitivity: the best within-source **PD** baseline drops from **0.9042** in the main partition to **0.7758** in partition 3, whereas **ALS** remains comparatively stable. The safest paper wording is therefore that the v4 conclusions are directionally informative but not fully insensitive to alternate near-optimal control splits.

1.9 Figure and table manifests

The manifest files below turn the notebook's selection decisions into explicit paper metadata. Legacy `step7/paper_fig*` image files are intentionally excluded; the manuscript should cite the stronger step-specific source figures directly.

```
[32]: def manifest_row(spec: dict) -> dict:
    pdf_path, png_path = figure_paths(spec["step"], spec["figure_stem"])
    return {
        **spec,
        "pdf_path": str(pdf_path.relative_to(REPO_ROOT)),
        "png_path": str(png_path.relative_to(REPO_ROOT)),
        "exists_pdf": pdf_path.exists(),
        "exists_png": png_path.exists(),
    }

figure_manifest_df = pd.DataFrame([manifest_row(spec) for spec in FIGURE_SPECS])

discovered_existing = []
for png_path in sorted(FIG_PNG_ROOT.glob("step*/*.png")):
    step = png_path.parent.name
    stem = png_path.stem
    if step == "step7":
        continue
    if any((row["step"] == step and row["figure_stem"] == stem) for row in
FIGURE_SPECS):
        continue
    pdf_path = FIG_PDF_ROOT / step / f"{stem}.pdf"
    discovered_existing.append({
        "tier": "supplement",
        "step": step,
        "figure_stem": stem,
        "recommended_caption": "Supplementary diagnostic figure retained for
repository depth.",
        "claim_supported": "Provides detailed backup for a step-specific
interpretation.",
        "why_included": "Non-redundant repository-facing detail.",
        "caveat": "Supplement only to avoid redundancy in the main paper.",
        "pdf_path": str(pdf_path.relative_to(REPO_ROOT)),
        "png_path": str(png_path.relative_to(REPO_ROOT)),
        "exists_pdf": pdf_path.exists(),
        "exists_png": png_path.exists(),
    })

if discovered_existing:
    figure_manifest_df = pd.concat([figure_manifest_df, pd.
DataFrame(discovered_existing)], ignore_index=True)
```

```

figure_manifest_df = figure_manifest_df.sort_values(["tier", "step",
↳ "figure_stem"]).reset_index(drop=True)
save_table_csv(figure_manifest_df, "paper_figure_manifest_v4.csv")
save_table_latex(
    figure_manifest_df,
    "paper_figure_manifest_v4.tex",
    "Recommended v4 paper-figure manifest.",
    "tab:paper_figure_manifest_v4",
    longtable=True,
)

all_csv_tables = sorted(path.name for path in TABLE_DIR.glob("*.csv"))
table_rows = list(TABLE_SPECS)
seen_tables = {spec["table_file"] for spec in TABLE_SPECS}
for table_name in all_csv_tables:
    if table_name in seen_tables or table_name in LEGACY_DUPLICATE_TABLES:
        continue
    if table_name.startswith("paper_"):
        continue
    table_rows.append({
        "tier": "supplement",
        "table_file": table_name,
        "purpose": "Supplementary numerical detail retained for repository and
↳ appendix use.",
        "main_columns": "See file columns.",
        "paper_use": "Supplement or appendix only.",
        "caveat": "Not every supplementary table should appear in the main
↳ manuscript.",
    })

table_manifest_df = pd.DataFrame(table_rows).sort_values(["tier",
↳ "table_file"]).reset_index(drop=True)
save_table_csv(table_manifest_df, "paper_table_manifest_v4.csv")
save_table_latex(
    table_manifest_df,
    "paper_table_manifest_v4.tex",
    "Recommended v4 paper-table manifest.",
    "tab:paper_table_manifest_v4",
    longtable=True,
)

display(figure_manifest_df[["tier", "step", "figure_stem", "exists_pdf",
↳ "exists_png"]])
display(table_manifest_df)

```

Saved table: report/tables/v4/paper_figure_manifest_v4.csv
 Saved table: report/tables/v4/paper_figure_manifest_v4.tex
 Saved table: report/tables/v4/paper_table_manifest_v4.csv
 Saved table: report/tables/v4/paper_table_manifest_v4.tex

	tier	step	figure_stem \
0	extended_main	step3	step3_bidirectional_asymmetry
1	extended_main	step3	step3_direction_intervals
2	extended_main	step4	step4_family_delta_j_consensus
3	extended_main	step4	step4_top_feature_recurrence
4	extended_main	step5	step5_corruption_cross
5	extended_main	step5	step5_feature_family_sensitivity
6	extended_main	step6	pca_scatter_by_condition
7	extended_main	step8	step8_best_transfer_leaders_heatmap
8	extended_main	step8	step8_within_source_baselines
9	main	step0	step0_authoritative_filtering_cascade
10	main	step1	step1_control_split_summary
11	main	step2	f1_within_condition_heatmap
12	main	step2	step2_subject_stride_leaders
13	main	step3	cross_condition_f1_heatmap
14	main	step3	degradation_heatmap
15	main	step3	step3_subject_confusion_matrices
16	main	step4	step4_delta_j_normalized_consensus_heatmap
17	main	step4	step4_delta_j_rf_heatmap
18	main	step4	step4_delta_j_spearman
19	main	step5	conformal_cross
20	main	step5	conformal_within
21	main	step5	noise_curves_cross
22	main	step5	step5_feature_sensitivity_cross
23	main	step5	step5_feature_sensitivity_within
24	main	step6	kmeans_scatter_k3
25	main	step8	step8_direction_degradation_ranges
26	main	step8	step8_sign_stability_matrix
27	supplement	step0	step0_raw_cohort_composition
28	supplement	step1	step1_feature_family_inventory
29	supplement	step2	step2_best_classifier_confusion_matrices
30	supplement	step2	step2_modal_frequency_heatmap
31	supplement	step2	step2_strategy_summary
32	supplement	step3	step3_best_direction_heatmap
33	supplement	step3	step3_per_subject_accuracy
34	supplement	step3	step3_within_vs_cross_by_direction
35	supplement	step4	delta_j_vs_degradation
36	supplement	step4	step4_dependence_proxies
37	supplement	step4	step4_dfa_alpha_stride_directionality
38	supplement	step4	step4_rf_within_vs_cross_importance
39	supplement	step4	waterfall_pd_to_hd_case_1
40	supplement	step4	waterfall_pd_to_hd_case_2
41	supplement	step4	waterfall_pd_to_hd_case_3

42	supplement	step5	noise_curves_within
43	supplement	step5	step5_corruption_within
44	supplement	step5	step5_cross_noise_variants
45	supplement	step5	step5_hard_label_consensus_curve
46	supplement	step5	step5_hardest_subjects
47	supplement	step5	step5_subject_sensitivity_cross
48	supplement	step5	step5_subject_sensitivity_within
49	supplement	step6	kmeans_elbow
50	supplement	step6	kmeans_silhouette
51	supplement	step6	pca_explained_variance
52	supplement	step8	step8_partition_overview_table

	exists_pdf	exists_png
0	True	True
1	True	True
2	True	True
3	True	True
4	True	True
5	True	True
6	True	True
7	True	True
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9	True	True
10	True	True
11	True	True
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13	True	True
14	True	True
15	True	True
16	True	True
17	True	True
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33	True	True
34	True	True

35	True	True
36	True	True
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39	True	True
40	True	True
41	True	True
42	True	True
43	True	True
44	True	True
45	True	True
46	True	True
47	True	True
48	True	True
49	True	True
50	True	True
51	True	True
52	True	True

	tier	table_file \
0	extended_main	step0_filter_flow.csv
1	extended_main	step1_control_partition_summary.csv
2	extended_main	step3_best_direction_recall_summary_v4.csv
3	extended_main	step5_feature_family_sensitivity_v4.csv
4	extended_main	step8_partition_overview_v4.csv
5	main	paper_main_results_summary_v4.csv
6	main	step2_subject_stride_leaders_v4.csv
7	main	step5_noise_robustness_summary_v4.csv
8	main	step8_direction_sensitivity_v4.csv
9	supplement	step0_raw_cohort_summary.csv
10	supplement	step1_feature_family_table.csv
11	supplement	step1_hash_summary.csv
12	supplement	step2_modal_params_v4.csv
13	supplement	step2_within_condition_summary_v4.csv
14	supplement	step3_bidirectional_asymmetry_v4.csv
15	supplement	step3_cross_condition_summary_v4.csv
16	supplement	step3_subject_accuracy_v4.csv
17	supplement	step3_subject_failure_summary_v4.csv
18	supplement	step4_delta_j_long_v4.csv
19	supplement	step4_delta_j_normalized_long_v4.csv
20	supplement	step4_delta_j_vs_degradation_v4.csv
21	supplement	step4_family_delta_j_long_v4.csv
22	supplement	step4_top_feature_recurrence_v4.csv
23	supplement	step4_top_features_v4.csv
24	supplement	step5_conformal_summary_v4.csv
25	supplement	step5_corruption_summary_v4.csv
26	supplement	step5_feature_top3_v4.csv
27	supplement	step5_noise_summary_v4.csv

28	supplement	step5_subject_sensitivity_v4.csv
29	supplement	step6_all_subjects_kmeans_metrics.csv
30	supplement	step6_all_subjects_pca_coordinates.csv
31	supplement	step6_binary_pathological_vs_control_kmeans_me...
32	supplement	step6_binary_pathological_vs_control_pca_coord...
33	supplement	step6_control_a_vs_control_b_kmeans_metrics.csv
34	supplement	step6_control_a_vs_control_b_pca_coordinates.csv
35	supplement	step6_disease_only_kmeans_metrics.csv
36	supplement	step6_disease_only_pca_coordinates.csv
37	supplement	step6_pca_feature_loadings.csv
38	supplement	step8_best_transfer_leaders_v4.csv
39	supplement	step8_within_baselines_v4.csv

purpose \

0	Cohort filtering stages and retained stride-ro...
1	Frozen Control A / Control B membership.
2	Best classifier per transfer direction with re...
3	Family-level permutation sensitivity across wi...
4	Partition-level Step 8 overview with overlap a...
5	Full direction-by-classifier cross-condition t...
6	Condition-level within-condition leaders with ...
7	Sigma-zero replay and key severity checkpoints...
8	Direction-level matched-degradation sensitivit...
9	Supplementary numerical detail retained for re...
10	Supplementary numerical detail retained for re...
11	Supplementary numerical detail retained for re...
12	Supplementary numerical detail retained for re...
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 37 Supplementary numerical detail retained for re...
 38 Supplementary numerical detail retained for re...
 39 Supplementary numerical detail retained for re...

	main_columns \
0	stage_label, rows
1	subject_id, group
2	Direction, Classifier, Subject F1, Delta subje...
3	scope, domain, classifier, family, drop
4	partition, role, overlap, independent_sensitiv...
5	Direction, Classifier, Subject F1, Within-sour...
6	Condition, Leader, Subject F1, Stride F1, Subj...
7	scope, classifier, f1@0, f1@0.10, f1@0.25, f1@...
8	Partition, Direction, Mean matched degradation...
9	See file columns.
10	See file columns.
11	See file columns.
12	See file columns.
13	See file columns.
14	See file columns.
15	See file columns.
16	See file columns.
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34	See file columns.
35	See file columns.
36	See file columns.
37	See file columns.
38	See file columns.
39	See file columns.

```

                                paper_use \
0      Methods transparency for cohort accounting.
1      Methods or supplement table for control-reuse ...
2      Compact Step 3 companion table if the manuscri...
3      Step 5 companion table for the feature-family ...
4      Supplementary context for the control-partitio...
5          Primary Step 3 numerical reporting table.
6              Exact Step 2 leader reporting.
7              Exact Step 5 checkpoint reporting.
8              Exact Step 8 sensitivity reporting.
9                  Supplement or appendix only.
10                 Supplement or appendix only.
11                 Supplement or appendix only.
12                 Supplement or appendix only.
13                 Supplement or appendix only.
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35                 Supplement or appendix only.
36                 Supplement or appendix only.
37                 Supplement or appendix only.
38                 Supplement or appendix only.
39                 Supplement or appendix only.

                                caveat
0      Descriptive only.
1      Important for protocol clarity but not a resul...
2      Best-direction rows do not summarize the full ...
3      Family averages compress heterogeneous directi...

```

4 Structural metadata rather than a headline res...
 5 Direction-level means are the safest headline ...
 6 Nested winners are protocol-specific rather th...
 7 This is a feature-space perturbation diagnosti...
 8 Sensitivity qualifies the main transfer story ...
 9 Not every supplementary table should appear in...
 10 Not every supplementary table should appear in...
 11 Not every supplementary table should appear in...
 12 Not every supplementary table should appear in...
 13 Not every supplementary table should appear in...
 14 Not every supplementary table should appear in...
 15 Not every supplementary table should appear in...
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 37 Not every supplementary table should appear in...
 38 Not every supplementary table should appear in...
 39 Not every supplementary table should appear in...

1.10 Final synthesis guidance for the manuscript

The curated figure and table set above supports a publication-facing story that stays close to the frozen v4 evidence.

- **Main claim:** zero-shot abnormal-vs-control transfer is measurable under leakage-controlled subject-level validation, but it is directionally asymmetric and usually degraded relative to within-source subject baselines.
- **Safe Step 3 wording:** direction-level mean subject-level matched degradation is positive across all six directions, while individual classifier-direction pairs still vary.
- **Safe Step 4 wording:** SHAP supports a diagnostic reliance-shift story centered on variabil-

ity, raw timing, and a small set of recurring timing-structure features; it does **not** establish causal biomarkers.

- **Safe Step 5 wording:** the robustness and conformal layers are diagnostic stress tests and exploratory uncertainty analyses, not raw-sensor realism or target-domain coverage guarantees.
- **Safe Step 6 wording:** PCA and K-means provide exploratory geometry context only.
- **Safe Step 8 wording:** alternate near-optimal control partitions preserve some conclusions but materially weaken others, because partition 3 reverses four of six direction-level matched-degradation signs.
- **Global caveat:** small subject counts remain the main limit on stronger inferential claims.

```
[33]: expected_step7_outputs = [  
    TABLE_DIR / "paper_figure_manifest_v4.csv",  
    TABLE_DIR / "paper_figure_manifest_v4.tex",  
    TABLE_DIR / "paper_table_manifest_v4.csv",  
    TABLE_DIR / "paper_table_manifest_v4.tex",  
    TABLE_DIR / "paper_main_results_summary_v4.csv",  
    TABLE_DIR / "paper_main_results_summary_v4.tex",  
]  
for path in expected_step7_outputs:  
    assert path.exists(), path  
  
for row in figure_manifest_df.itertuples():  
    if row.tier in {"main", "extended_main"}:  
        assert row.exists_pdf, row.pdf_path  
        assert row.exists_png, row.png_path  
  
assert all(spec["step"] != "step7" for spec in FIGURE_SPECS)  
  
print("PASS: Step 7 synthesis notebook curated existing v4 figures and wrote_  
↳the paper manifests.")
```

PASS: Step 7 synthesis notebook curated existing v4 figures and wrote the paper manifests.